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FAHIMEH ARMIN

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FAHIMEH ARMIN

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TABLE OF ABBREVIATIONS

AWG	Arrayed waveguide grating
CAGR	Compound annual growth rate
CMOS	Complementary metal-oxide-semiconductor
DFB	Distributed feedback
EDG	Echelle diffraction grating
EME	Eigen mode expansion
FDE	Finite difference eigenmode
FDTD	Finite difference time domain
FSR	Free spectral range
LPCVD	Low-pressure chemical vapor deposition
MMI	Multi mode interferometer
MZI	Mach-Zehnder interferometers
NG-PON	Next generation passive optical networks
PDK	Process design kit
PECVD	Plasma-enhanced chemical vapor deposition
PIC	Photonic integrated circuit
QW	Quantum well
SiN	Silicon nitride
SLM	Sagnac loop mirror
SOI	Silicon on insulator
TE	Transverse electric
TM	Transverse magnetic
WDM	Wavelength division multiplexing

RÉSUMÉ

Les plates-formes photoniques de silicium sur isolant (SOI) sont constituées d'une couche de silicium cristallin sur une couche tampon de dioxyde de silicium ($SiO_{2)}$. Le contraste d'indice de réfraction élevé entre le silicium et le SiO_2 permet des courbes de guide d'onde optiques à faible rayon et donc des dispositifs compacts. Néanmoins, ce contraste d'indice élevé entraîne des pertes de diffusion importantes. Une alternative importante au silicium est le nitrure de silicium (SiN). Le SiN est également un matériau courant dans les processus de fabrication CMOS (complementary metal-oxide-semiconductor) et offre un contraste d'indice de réfraction relativement élevé pour créer des guides d'ondes optiques. Les circuits photoniques en SiN utilisent la même couche tampon, le même substrat et le même revêtement que les plates-formes SOI. De plus, les pertes de propagation des guides d'ondes en SiN sont inférieures d'un ordre de grandeur à celles des guides d'ondes en silicium, car leur contraste d'indice de réfraction plus faible les rend moins sensibles à la rugosité et offre des tolérances plus élevées aux variations de fabrication.

Le contraste d'indice de réfraction plus faible entre le SiN et le SiO_2 crée d'autres défis dans les dispositifs intégrés. Contrairement à la photonique sur silicium, les guides d'ondes sur les plates-formes SiN ne peuvent pas avoir de courbures prononcées. Le mode fondamental est moins confiné dans le noyau du guide d'ondes et donc un guide d'ondes en SiN courbé avec un petit rayon rayonne une partie importante de la puissance optique. Pour éviter ces pertes, les guides d'ondes en SiN sont limités à de grands rayons de courbure, ce qui se traduit par des dispositifs ayant un grand encombrement. Cela réduit la densité d'intégration réalisable dans les plates-formes SiN par rapport à la photonique sur silicium, ce qui peut avoir un impact sur la taille et le coût des systèmes optiques.

Cette étude explore des méthodes pour remédier aux manques de la photonique SiN tout en tirant parti de ses faibles pertes de propagation pour créer de nouveaux filtres optiques, qui peuvent être utilisés comme éléments constitutifs de futurs réseaux optiques élastiques. Les composants proposés sont compatibles avec les procédés de fabrication en fonderie pour produire des systèmes rentables. Nous avons étudié les coupleurs de 3 dB multimode qui sont utilisés dans un nouvel interféromètre compact avec un encombrement 50 fois plus petit que les conceptions conventionnelles. Il a été démontré qu'en utilisant des éléments de chauffage locaux, le dispositif peut être

facilement réglé pour compenser les erreurs de fabrication. Bien que l'utilisation de cet interféromètre dans l'une de nos configurations de filtre proposées dans le cadre d'un réflecteur puisse en réduire la surface, l'espacement des canaux du filtre est toujours limité par la longueur totale des guides d'ondes. Le remplacement de l'interféromètre et de la structure réfléchissante par notre réseau de Bragg circulaire optimisée augmentera l'espacement possible entre les canaux à plus de 1000 GHz. Dans un système a multiplexage en longueur d'onde, cela signifie que plus de canaux peuvent être utilisés et qu'il y aura moins d'interférences entre les canaux adjacents. Le réseau proposé a une bande passante de 1 et 3 dB de 130 et 195 nm, respectivement, et offre une réduction de taille jusqu'à 60 fois par rapport aux miroirs en boucle de Sagnac. Une telle bande passante permet à l'appareil de fonctionner sur toute la bande optique utilisée en télécommunication, même en cas de léger décalage de la longueur d'onde de conception centrale dû à des erreurs de fabrication.

Mots clés : Plateforme en nitrure de silicium, Coupleurs d'interférences multimodes, Cavités de Fabry-Perot, Filtres optiques intégrés

ABSTRACT

Silicon-on-insulator (SOI) photonic platforms consist of a crystalline silicon layer over a silicon dioxide (SiO_2) buffer layer. The high refractive index contrast between silicon and SiO_2 allows for optical waveguide bends with small radius and therefore compact devices. Nevertheless, the high index contrast leads to significant scattering losses at nm-scale roughness. One important alternative to silicon is silicon nitride (SiN). SiN is also a common material in CMOS fabrication processes and offers a relatively high refractive index contrast to create optical waveguides. SiN photonic circuits use the same buffer layer, substrate, and cladding as SOI platforms. Moreover, the propagation losses of SiN waveguides are an order of magnitude lower than those of silicon waveguides, since their lower refractive index contrast makes them less sensitive to roughness scattering and provide higher tolerances to fabrication variations.

The lower refractive index contrast between SiN and SiO_2 creates other challenges in integrated devices. Unlike silicon photonics, waveguides on SiN platforms cannot have sharp bends. The fundamental mode is less confined in the waveguide core and thus a SiN bent waveguide with a small radius radiates a significant portion of the optical power. To avoid bending losses, SiN waveguides are limited to large radius of curvatures, which in turn results in devices with a large footprint. This reduces the integration density achievable in SiN platforms in comparison to silicon photonics, which can impact the size and cost of optical systems.

This study investigates methods to address the short coming of SiN photonics while taking advantage of its low propagation losses to create novel optical filters, which can be used as the building blocks for future elastic optical networks. The proposed components are compatible with foundry-based fabrication processes for cost effective systems. We investigated 3 dB MMI couplers that are used in a novel compact interferometer with a footprint 50 times smaller than the conventional designs. It has been shown that using local heaters the device can be easily tuned to compensate fabrication errors. Although using this interferometer in one of our proposed filter configurations as a part of a reflector can decrease floor space, the channel spacing of the filter is still limited by the total length of the waveguides. Replacing the interferometer and the reflecting structure with our optimized circular Bragg grating will increase the channel spacing to more than 1000 GHz. In a WDM system this means that more channels can be used and that there will be less interference between the

adjacent channels. The proposed grating has a 1- and 3-dB bandwidth of 130 and 195 nm, respectively, and provides a size reduction up to 60 times in comparison to Sagnac loop mirrors. Such a large reflection bandwidth allows the device to operate over the entire optical band even in the case of slight shift form the center design wavelength in case of fabrication errors.

Keywords: Silicon nitride platform, Multimode interference couplers, Fabry-Perot cavities, Integrated optical filters

CHAPTER I

INTRODUCTION

During the past decades increasing demand in digital services, social media, and cloud computing have led to higher bandwidth and larger capacity requirements. This obligation has equally affected the private, public, and business data centers which now must handle an ever-increasing demand for data. Figure 1-1 shows a clear representation of this growth by comparing the network traffics in different regions over the years (Kish *et al*. 2018; Winzer and Neilson 2017). The average traffic passing through the Amsterdam internet exchange (AMS -IX) between 2001 and 2017 is shown in Figure 1-1 (a) revealing a 60% increase in the compound annual growth rate[1] (CAGR) in total. Similar expanding rates are observed for the Seattle (Figure 1-1 (b)) and the UK (Figure 1-1 (c)) exchange centers in only ten years from 2008 to 2017. More detailed observations are presented in Figure 1-1 (d) – (f) by plotting the total broadband downloads in Japan, Global mobile, and video traffic, respectively. Only Netflix streaming had a constant growth rate of 50% from 2012 to 2017, competing with the 70% growth of YouTube uploads. According to the 2020 Forecast Highlights reported by CISCO ("Cisco Annual Internet Report," n.d.), video traffic grew 4 folds from 2015 to 2020 with an annual rate of 31%. As stated in this report, the annual

[1] CAGR is a measure of investment grow over a specific number of years and corresponds to the average return of investment funds ("Compound Annual Growth Rate" 2021).

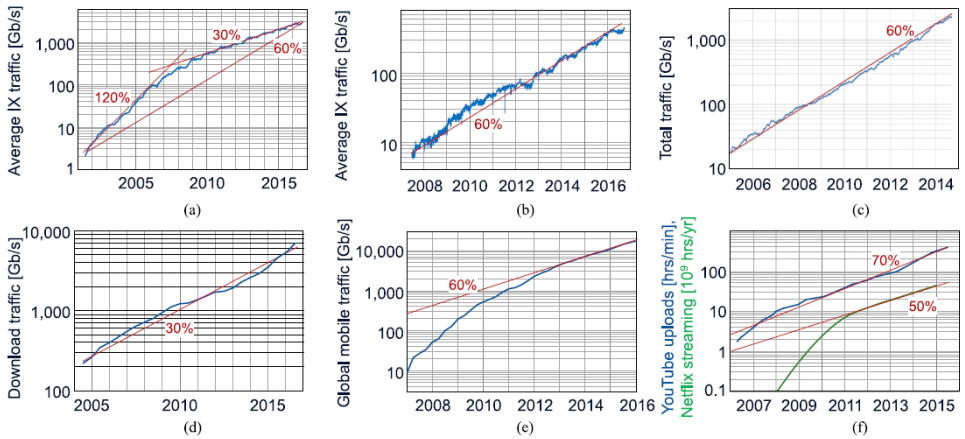


Figure 1-1: Growth of the network traffic in different regions, (a) Amsterdam, (b) Seattle, (c) Total demand in UK, (d) Total download traffic in Japan, (e) Global mobile data traffic and (f) two examples of video traffic (Winzer and Neilson 2017).

growth rate is 53% for mobile data, 46% for gaming, and 26% for internet protocol (IP) videos.

The substitution of copper lines with fiber optics has mitigated the data and bandwidth issues to a great extent. These improvements started in the early 1970s with the introduction of optical fiber technology and then were followed by developments such as increases in the bit rates of single wavelength channels and later by shifting the transmission windows from 800 to 1550 nm.

Recently silicon photonics structures based on silicon-on-insulator platforms have been studied extensively to develop integrated devices. The main aim behind integrated photonic circuits is to reduce the total system cost. Silicon photonics is considered as being low cost and high yield from an industrial perspective because it leverages mature metal oxide semiconductor manufacturing (Jalali and Fathpour 2006).

1.1 Thesis Objectives

Next generation passive optical networks known as NG-PON2 promises faster and more reliable data traffic. Although it seems thrilling it is not without its challenges. This work is part of a collaborative research program with compony Aeponyx with the main purpose of developing a new microfabrication platform for integrating microelectromechanical actuators and optical waveguides based on silicon nitride. Accordingly the objective of this thesis is to design, analyze, fabricate, and characterize novel optical filters based on silicon nitride (SiN) waveguides. The study aims to create a filter with a wide free spectral range (FSR) while respecting the fabrication criteria and limitations. As a part of a WDM network, such filters can isolate channels with strong adjacent channel rejection while maintaining the transmitted signal quality. The filter response is based on the multiple interference occurring in a Fabry-Perot cavity that is formed by connecting two reflectors. Two different reflection mechanisms are used here which are Mach- Zehnder assisted Sagnac loops and Bragg gratings. A part of this thesis is dedicated to the analysis, design and characterizations different components and devices of that are used in these filter designs.

1.2 Methodology

As the first step in every research, we started with an extensive literature study on passive optical networks, different configurations of filters and their components, and fabrication methods and processes. The proposed designs use a SiN platform on a 200 mm Silicon (Si) wafer with a CMOS compatible fabrication process. The material stack in all designs consists of a 440 nm thick SiN waveguide core on a 3.2 μm bottom cladding, and 3.2 μm of top claddings of SiO_2. The claddings are made with low

pressure chemical deposition (LPCVD) of tetraethoxysilane (TEOS) while the SiN waveguide layer is silicon rich LPCVD.

The filter responses are calculated using the transfer matrix method implemented in MATLAB. Thus, the role of each component and parameter can be analytically studied. The filter can also be implemented in Lumerical INTERCONNECT while using ideal model of the components. However, to design and optimize the different components and to have a more realistic compact model, we have simulated the structures with different numerical techniques such as Finite Difference Eigenmode (FDE) solver, Eigen-mode expansion (EME) method and the finite difference time domain (FDTD) solvers. All these numerical solvers, as well as INTERCONNECT solver for integrated photonic circuits, are parts of Ansys Lumerical photonics simulations and design software.

After designs and optimizations, masks are prepared for fabrication in L-edit software, respecting the layer specifications and PDKs. Fabrications are conducted at the C2MI by AEPONYX and at INRS. The measurements are performed at AEPONYX and UQAM lab facilities. The measurement results from these fabrications are compared to the simulation and analytical results to complete the loop of verification and improvement.

1.3 Thesis Contributions

In this thesis, we have designed three different filter configurations, contributing to the field of integrated photonics. The first configuration is a fabrication tolerant comb filter based on a Fabry-Perot cavity. Two Signac loop mirrors each fed through an

unbalanced Mach-Zehnder interferometer create a resonant cavity. Using a large bending radius for the loop mirrors and a novel configuration for the interferometers that minimize the length of their arms and enables the implementation of a small length difference results in lower losses and footprint which could reach higher FSR and quality factor. The concept of this filter was presented at the 2020 OSA advanced photonics congress (13 – 16 July 2020) (Armin, Nabki, and Ménard 2020).

The second filter configuration also uses a MZI assisted Sagnac loop as the reflection mechanism. However, for this filter, we have designed, analysed, and characterized a novel interferometer which helps in reducing the device footprints and increasing the integration density. This interferometer uses waveguides with different effective indexes instead of lengths to avoid waveguide bends. This novel interferometer was first presented at the 2021 Frontiers in Optics and Laser Science conference (1 – 4 November 2021) (Armin, Nabki, and Ménard 2021a). An extended study including the thermal tuning analysis and experimental results of a mirror using this interferometer has been submitted to Optica Optics Express journal.

The third filter configuration uses compact SiN Brag grating reflectors which are designed to maintain a wide reflection band with low insertion loss. Replacing the MZI assisted Sagnac loops with this design will significantly reduce the both the cavity length and the physical dimensions of the filter. This part of our studies has been presented at the 2021 IEEE Group Four Photonic Conference (7-10 December 2021) (Armin, Nabki, and Ménard 2021b).

The following list presents the full thesis contributions in brief:

- Design, simulation, and characterization of a novel ultra-compact MZI in SiN using variations in effective index instead of length;
- Design, analysis, and implementation of a novel compact MZI assisted tunable reflector;
- Design and optimization of the first SiN compact broadband circular Bragg grating reflector;
- Design and characterization of a novel bent MZI for compact devices;
- Implementation and analysis of the first SiN FP filter using Sagnac loop mirrors designed for circuits with high integration density and able to provide high quality factors;
- And the analysis and simulation the first SiN FP filter using compact broadband Bragg grating mirrors and able to provide a FSR of up to 1000 GHz.

List of Publications:

1. Armin, Fahimeh, Frederic Nabki, and Michaël Ménard. 2020. "Fabrication Tolerant Compact Tunable Silicon Nitride Comb Filters." In *OSA Advanced Photonics Congress (AP) 2020 Paper ITu4A.7*, ITu4A.7. Optical Society of America.
2. Armin, Fahimeh, Frederic Nabki, 2021. "Compact Silicon Nitride Interferometer." In *Frontiers in Optics + Laser Science 2021 (2021), Paper FTh6B.5*, FTh6B.5. Optica Publishing Group.
3. Armin, Fahimeh, Frederic Nabki, 2021. "Compact Silicon Nitride Circular Grating Reflectors." In *2021 IEEE 17th International Conference on Group IV Photonics (GFP)*, 1–2.

1.4 Thesis Organization

After a brief introduction presented in this chapter, chapter 2 of this thesis starts with a short history of integrated photonic systems and their evolution, and then continues with introducing some important aspects and theories of these systems. Our proposed filter configurations are presented in chapter 3 where the design and fabrication challenges are also discussed. Multimode interference couplers, Mach-Zehnder interferometers, and reflectors are the key components in the proposed designs and therefore, chapters 4, 5 and 6 are each dedicated to one of these elements. Chapter 4 presents the simulation and fabrication results of couplers which are used in the final filter design. Chapter 5 reviews the theory, design process, and characterization of a novel compact interferometer. This chapter also discusses the tunabilty of the proposed design using thermo-optical effects. Sagnac loop mirrors and integrated Bragg gratings are studied in chapter 6 as two different reflectors. Bragg gratings are introduced as a compact alternative to Sagnac loop mirrors in one of our proposed filter designs. As the last chapter, Chapter 7 is dedicated to an extensive study of different filter configurations. It starts with discussing the two different reflecting mechanism, then moves on to the analytical study of the filters by calculating the transfer functions of each configuration. This chapter ends with 3 different sections that study each proposed design separately.

CHAPTER II

INTEGRATED PHOTONICS SYSTEMS

In the late 1990s, the development of optical amplifiers that enable wavelength division multiplexing (WDM) significantly increased the capacity of optical telecommunication systems (Essiambre and Tkach 2012). By using WDM, multiple channels can be encoded as different wavelengths in a single fiber. This type of multiplexing and the technology developed based on it improved system capacity from less than 100 MB/s up to 1 Tb/s by the year 2000 (Winzer, Neilson, and Chraplyvy 2018). What also pushed the boundaries further was the reduction of losses in optical systems. Although optical fibers do not exhibit ohmic losses, they still suffer losses due to scattering and absorptions. The effect of these losses was significantly mitigated through innovative solutions presented during these years (Essiambre and Tkach 2012).

Despite all these improvements, high-speed communication networks still lacked an efficient photonics/microelectronic interface. By implementing new technologies and moving toward smaller-sized transistors, it is now possible to miniaturize and integrate many transistors on a single chip. However, the downside to this intensive integration is that heat dissipation becomes an issue. In addition to that, the reduction in the width of the interconnects results in higher resistivity and thus resistor-capacitor delays (Doylend and Knights 2012). One solution to this problem was the parallelization of processors which was also limited by the speed of communication between the processing cores. Optical interconnects emerge as a promising solution

that offers lower energy consumption, a smaller footprint, and faster interconnects (R. Soref 2006; Winzer and Neilson 2017).

Figure 2-1 (a) shows a cross-section view of the optical waveguide proposed by Soref and Lorenzo in 1985 (R. A. Soref and Lorenzo 1985). By adding a high resistivity epitaxial silicon layer on a heavily doped monocrystalline silicon layer, they presented the first all-silicon optical component. They reported an index contrast of 10^{-3} to 10^{-1} between the channel and the substrate. These numbers would seem insignificant in comparison to the silicon-on-insulator (SOI) platform, but it was a breakthrough at the time. The SOI platform was first theoretically introduced by Kurdi and Hall in which they show that the new waveguides can reach 1dB/cm propagation loss for the fundamental TE mode at a wavelength of 1300 nm (Kurdi and Hall 1988). Based on their studies, in 1991, Schmidtchen *et al.* fabricated the first single-mode SOI waveguide. In their fabricated structure, as it is illustrated in Figure 2-1 (b), a 400 nm thick silicon dioxide layer separated the epitaxial silicon rib waveguide from the silicon substrate (Schmidtchen *et al*. 1991).

Schmidtchen *et al.* reported a propagation loss below 0.5 dB/cm for the fundamental TE mode in waveguides with widths between 3 and 12 μm. These numbers are crucial for further integration and miniaturization of devices. The low

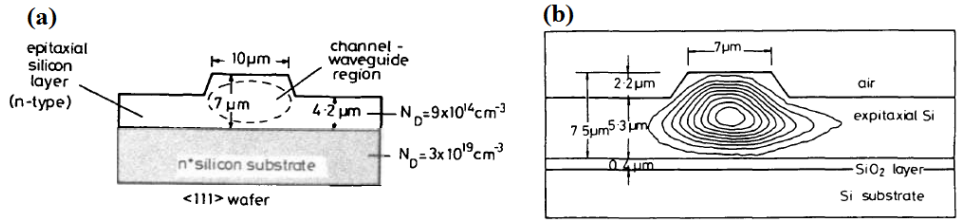


Figure 2-1: Cross sectional view of first optical waveguides, (a) silicon on doped silicon (R. A. Soref and Lorenzo 1985), and (b) Silicon on silicon dioxide (Soref 1993).

propagation loss reported is mainly due to the high index contrast of silicon (3.48 at λ=1550 nm) with the claddings (1.45 for silicon dioxide or 1 for air). With these promising results, a year later the concept of "superchip" was proposed in which a single silicon die would be used for integration of both optical and electronics circuits (Abstreiter 1992; R.A. Soref 1993). This superchip is schematically presented in Figure 2-2. Although this schematic includes all the building blocks, light source, modulators, and amplifiers could not be realized within a silicon chip at the time. This limitation is due to the indirect bandgap of silicon and forced the designers to continue with external sources, amplification, and modulation.

The full vision of the superchip could not be realized at this time, but the building blocks of silicon photonics continue to evolve. At the first International Conference on Group IV Photonics, it was shown that many research teams and companies had already started investing in the study and development of waveguides, couplers, resonators, and other basic building blocks as well as components with more complex functions such as filters and multiplexers. All these efforts were to move toward an efficient photonic integrated circuit (PIC). In parallel, many works were also published on the ways to integrate sources, detectors, and modulators.

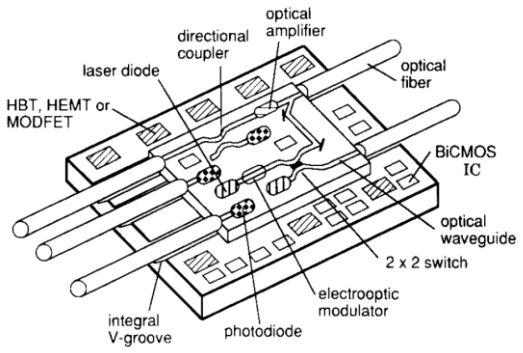


Figure 2-2: Visualization of a "Superchip" as it was presented by Soref in 1993 (Abstreiter 1992).

Table 2-1, which is adapted from (Kish *et al*. 2018), shows the trend and evaluation of key devices of PICs starting with an indium antimonide (InSb) photodetector in 1959. At the end of the list stand the modulators and arrayed waveguide gratings (AWGs) as the components used to facilitate the implementation of filters and commercial WDM systems in 1999 (Alegria *et al*. 1999).

Table 2-1: Developments of key devices of PIC (Kish *et al*. 2018)

Year	Technology Break-Through
1959	III-V photodetector
1966	III-V electro-absorption (EA) modulator
1971	DFB laser proposal
1974	DFB injection laser demonstration
1975	DFB injection laser (Continues wave at room temperature)
1975	Self-imaging in planar optical waveguides
1977	Quantum well (QW) injection laser
1979	QW MOVPE injection lasers (Continues wave at room temperature)
1978	InGaAsP/InP photodetectors
1981	InP-based DFB injection laser (Continues wave at room temperature)
1984	Strained QW injection laser
1984	III-V Mach-Zehnder modulator
1984	InP-based electro-absorption (EA) modulator
1986	Strained QW laser band-structure engineering
1986	Waveguide photodetector
1988	Phased array waveguides
1989	InP-based Mach-Zehnder modulator
1991	Arrayed waveguide gratings (AWGs)
1992	III-V AWG devices

2.1 Integrated Photonics Materials

The term silicon photonics itself is enough to understand the role of silicon in integrated photonics systems. Today, silicon-on-insulator (SOI) is the main platform for silicon photonic devices. Two main features are responsible for this interest in silicon as the main waveguide material in photonics. The first one is the mature fabrication infrastructure developed for CMOS technologies. Not only different volumes of silicon photonic chips can be fabricated using the existing technology, but prototyping is also executed in multi-project wafers at very low costs. Low cost and matured technology reduced the fabrication risks and allowed for more energy invested in innovative designs (Rahim *et al*. 2017).

The second reason for the wide adoption of silicon is its high refractive index contrast with the claddings which ensures maximum confinement. The advantage of this confinement is a reduction in footprint while fabricating complex devices and therefore, more functions can be integrated on a single chip.

The SOI platform consists of a crystalline silicon layer placed on a silicon oxide buffer layer. Commercial wafers with various silicon thicknesses are produced for the fabrication of waveguide-based devices (Baets *et al*. 2016). Although a thickness of 220 nm is often used in industry, higher thicknesses such as 310 and 400 nm are also reported in the literature. Thicker waveguides facilitate the integration of laser sources. With conventional waveguide dimensions, which are 220 ×500 nm, the high index contrast allows for bends with a radius as small as 5 μm at a wavelength of 1500 nm (Chrostowski and Hochberg 2015). Nevertheless, the high index contrast results in more scattering loss at nm-scale roughnesses.

The evaluation of alternative materials was always a part of research in photonics. The reason for this interest is the desire to work at wavelengths where silicon is not transparent. Silicon nitride is the main substitution material for silicon. It is also a common material in CMOS fabrication and presents a relatively good index contrast. For silicon nitride chip fabrication, the same buffer layer, substrate, and cladding as SOI platforms are used.

There are two main deposition techniques for silicon nitride: Low-pressure chemical vapor deposition (LPCVD) at high temperature or plasma-enhanced chemical vapor deposition (PECVD) at lower temperature. In the LPCVD technique, thicker layers of silicon nitride are deposited by increasing the silicon concentration or by using a thermal cycling process. There is also better control over the homogeneity of the effective index and layer thickness then with PECVD. In the PECVD technique, depending on the deposition condition, the refractive index of silicon nitride can be altered between a relatively lower (less than 1.7) and higher values (more than 2) (de Ridder *et al*. 1998). The first technique is common for devices working at telecommunication wavelengths. Depositing at high temperatures allows to drive out the hydrogen but at the same time, co-integration with CMOS is challenging (Rahim *et al*. 2017). A comparison of the refractive indices and absorption coefficient of silicon nitride in both techniques is presented in Figure 2-3.

The transparency of silicon nitride is maintained over a wide range in the visible region while silicon has high absorption at wavelengths lower than 1100 nm. Also, the propagation losses of silicon waveguides are an order of magnitude higher than silicon nitride since the lower refractive index contrast of silicon nitride waveguides makes them more tolerant to roughness scattering.

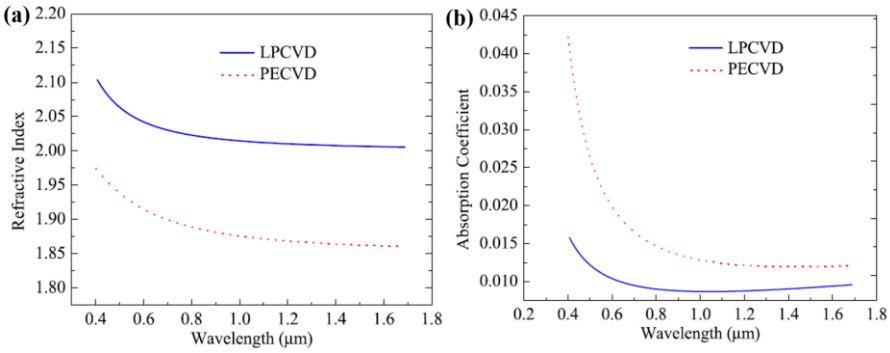


Figure 2-3: (a) Refractive index and (a) absorption coefficient of silicon nitride from two different deposition methods (de Ridder *et al.* 1998).

Finally, silicon has a great Kerr nonlinearity, but two-photon absorption makes it impractical since it imposes extra losses at telecommunications wavelengths. In silicon nitride, on the other hand, the Kerr effect is weaker, but the two-photon absorption phenomenon is zero due to its large bandgap,

2.2 Mode Propagations in Dielectric Waveguides

The routing of optical energy in integrated photonic devices is realized through dielectric waveguides. Depending on the function and wavelength, optical waveguides take many forms from metallic slabs to glass fibers. The most common type of waveguide in photonic integrated circuits (PICs) is the strip waveguide which consists of a high index dielectric with a rectangular cross section surrounded by a lower index cladding. This section is dedicated to waveguides since they are basic components of PICs.

The structure of a waveguide consists of a high refractive index core surrounded by a lower index cladding, as is shown in Figure 2-4. Because of this contrast in the

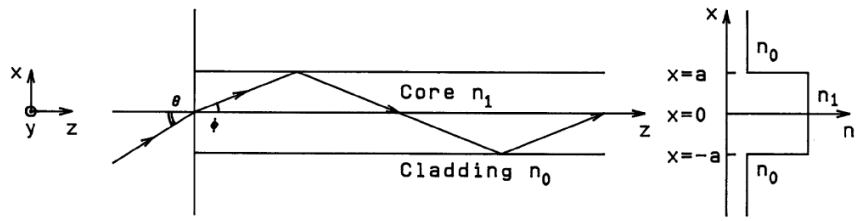


Figure 2-4: Schematic presentation of an optical waveguide and its refractive index profile (Okamoto 2005).

refractive indices, total internal reflection occurs at the interface of the core and cladding and thus the light that is coupled to a waveguide is confined and propagates inside the core (Okamoto 2005). From an electromagnetic point of view, the optical modes of a waveguide are obtained by solving the Helmholtz wave equations (Saleh and Teich 2007) for the transverse electric (TE) and transverse magnetic (TM) waves. In photonic circuits, single-mode optical waveguides are of interest. Thus, the dimensions of the waveguide are chosen so that it can only support the fundamental modes, namely TE_0 or TM_0.

The calculated field profile of the fundamental TE mode is shown in Figure 2-5 for two different core materials at the wavelength of 1550 nm. Both silicon (Si) and silicon nitride (Si_3N_4) are widely used in photonic integrated circuits. At 1550 nm the refractive index of silicon is 3.48 whereas the refractive index of silicon nitride is 2.03. Thus, the optical mode in silicon nitride is less confined to the waveguide core in comparison to silicon due to the smaller index contrast with the silicon dioxide cladding. The dimensions that are shown in Figure 2-5 are typical dimensions for a single-mode waveguide using silicon or silicon nitride core in C and L bands which ranges from 1520 to 1620 nm.

The propagation constant of the mode in the propagation direction (*z*-direction in the structure shown in Figure 2-5), β can be written as:

$$\beta = k_0 n_{eff} \,, \tag{2-1}$$

where $k_0=2\pi/\lambda_0$ is the propagation constant in free space with λ_0 being the free-space wavelength. Each optical mode is characterized by its effective refractive index, n_{eff}.

For both silicon and silicon nitride waveguide cores, n_{eff} is a complex number and its real part represents the ratio of the speed of light in free space (c = 3 $\times 10^8$ m/s) to the speed of a wave inside the waveguide for the specific mode. The imaginary part, on the other hand, corresponds to the losses in the propagation medium. However,

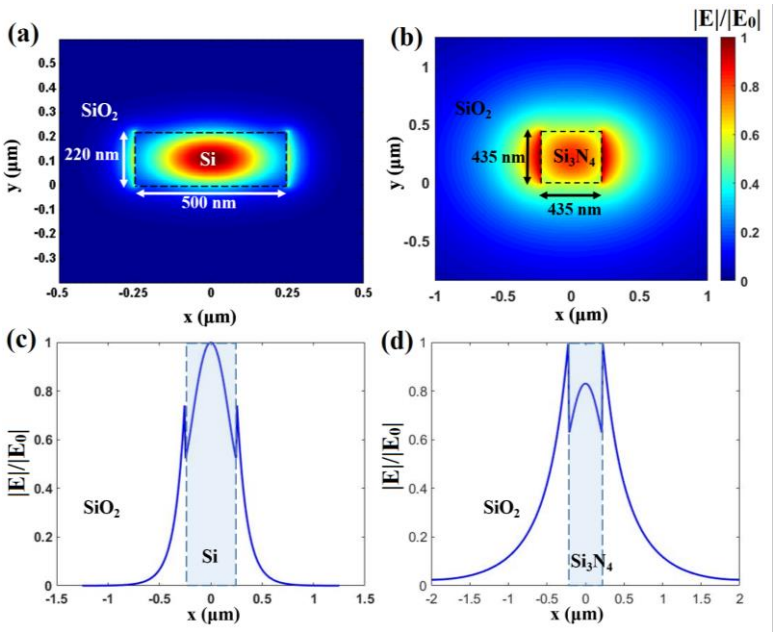


Figure 2-5: Profile of the electric Field of the fundamental TE mode in (a) silicon waveguide in silicon dioxide cladding and (b) silicon nitride waveguide with silicon dioxide cladding. (c) Normalized electric field at the cross section of the silicon core and (d) silicon nitride core.

optical losses are mainly due to scattering from the roughness of the waveguide sidewalls which is created during the lithography and etching processes. The power lost due to scattering depends on the mode shape in the core and the amount of light at the waveguide edges. Typical losses in silicon waveguides are reported to be less than 3 dB/cm (Chrostowski and Hochberg 2015). For straight waveguides, other sources of loss, such as material and leakage losses are usually not significant.

The effective refractive index corresponds to the phase velocity of light by showing the ratio of the speed of light in the free space to the speed in the propagation medium. The group index, n_g, on the other hand, represents the group velocity, the speed at which the pulse propagates. The group index can be related to the effective refractive index through (Chrostowski and Hochberg 2015)

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \tag{2-2}$$

Figure 2-6 compares the effective index and group index values for silicon and silicon nitride waveguides. The waveguides have the dimensions shown in Figure 2-5.

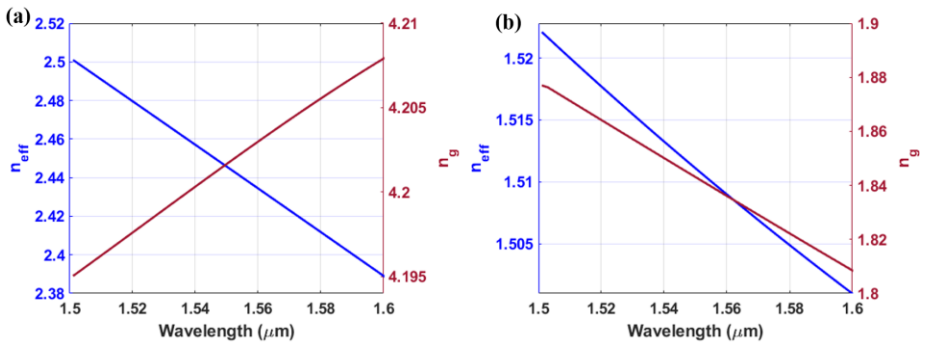


Figure 2-6: Effective refractive index and group index of the fundamental TE mode versus wavelength for (a) silicon and (b) silicon nitride waveguides

At each specific wavelength, the effective and group indexes depend on the geometry of the waveguide. A square waveguide has the advantage of being polarization insensitive, which means that the effective and group index of the TE and TM modes are similar in this waveguide as a result of its symmetry.

In component design as well as signal routing on a chip, it is needed to have bent waveguides. But the modal mismatch between the bent and the straight waveguide results in excessive losses. One common method to overcome this is to use relatively large bend radiuses so that the change is gradual in the propagation medium.

2.3 Couplers

Routing the optical wave from one channel to another or splitting it between different waveguides as well as combining waves from different waveguides is commonly required in photonic circuits. Splitting and combining waves in an optical system is realized through couplers. Here we study two common types of couplers that are based on two entirely different concepts. We start with directional couplers that are schematically presented in Figure 2-7.

Inside the waveguide core light cannot be entirely confined (Figure 2-5 (c) and (d))

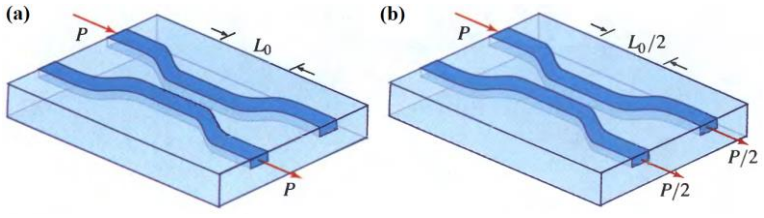


Figure 2-7: Schematic presentation of a directional coupler as (a) route the light into another waveguide and (b) split the input power equally between two waveguides (Saleh and Teich 2007)

and a part of it evanescently decays outside the waveguide. If another parallel waveguide is located close enough, this evanescence tail can couple the mode energy totally or partially into the second waveguide. Thus, a determining factor in the coupling between the waveguides is the gap between them.

Another important factor is the coupling length, shown in Figure 2-7 by L_0. This is the effective length at which the light can entirely couple into the second waveguide with similar dimensions. By using half of this length as in Figure 2-7 (b), the component will be a 3 dB coupler by splitting the input power equally between the two output waveguides. Using the same trend in design, arbitrary coupling rations can be obtained.

Another type of coupler is based on image theory (Bryngdahl 1973) and can effectively divide the input power between multiple outputs. These couplers are known as multi-mode interference (MMI) couplers since, as shown in Figure 2-8, the input wave is coupled to a relatively wide waveguide that supports many modes. Upon interference between these modes, multiple images of the input mode are created. The

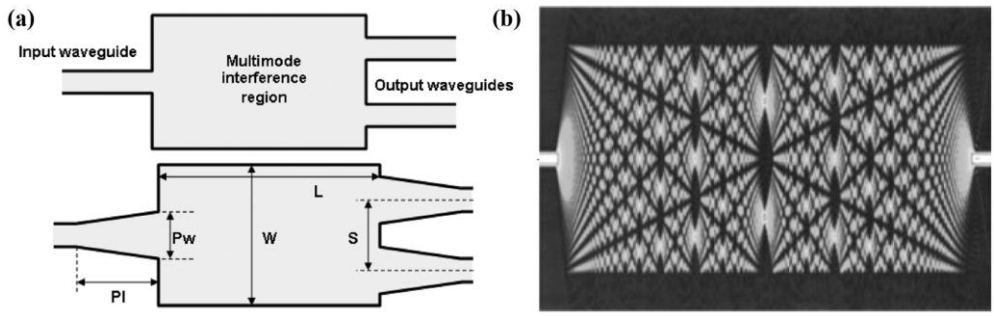


Figure 2-8: (a) Schematic presentation of a MMI coupler with and without tapered input and output waveguides (Thomson *et al*. 2010). (b) Electric field profile of an MMI coupler showing image formation at different locations in the MMI region (Bryngdahl 1973).

modal mismatch between the MMI region and the input or output waveguides is the main source of loss in these components. One solution to minimize loss is to use a tapered waveguide as in Figure 2-8 (a) (Thomson *et al*. 2010). Image formation inside the MMI region is presented in Figure 2-8 (b) and shows how the different sets of the input image are formed at different locations.

Both directional and MMI couplers are widely used in photonic circuits. However, directional couplers are more sensitive to thermal and fabrication variations that commonly occurs. MMI couplers, on the other hand, are less sensitive to changes in dimensions and their splitting ratio is less sensitive to temperature changes. As mentioned before, the modal mismatch, and thus the insertion loss of MMI couplers is one major drawback for this type of couplers.

Figure 2-9 demonstrates the effect of fabrication errors on these two types of couplers. The inset shows the electric field in the components. Simulations have been done at a wavelength of 1550 nm for silicon nitride as the core material. Both couplers have the same length. The gap between the two waveguides and the output separation

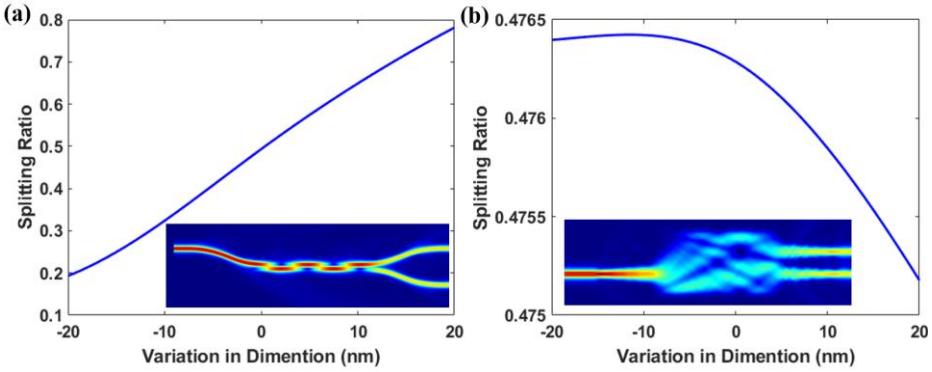


Figure 2-9: Splitting ratio of the couplers as the function of change in their dimensions for (a) directional coupler and (b) MMI coupler.

of the MMI coupler have been changed by ±20 nm. As can be seen, the MMI output hardly changes in this range whereas the output of the directional coupler changes severely even with a slight variation.

Another component used as a coupler is called a Y-branch, a Y-splitter, or sometimes a T-coupler (Saleh and Teich 2007). It got its name from its Y shape structure, and it is a simple structure to split the input power into two separate waveguides or reversely, combine the optical wave from two different waveguides in one output port. The schematic of a Y-splitter is shown in Figure 2-10 (a).

As a splitter, the input intensity is equally routed to two branches, but the same trend applies when it is used as a combiner. This means the input intensity in each branch is divided into two modes. These modes are the fundamental mode and second-order mode which in this case is a radiation mode. Therefore, while using the y-splitter in combiner mode, if light is only coupled in one port, the output will be equal to half of the input intensity (Figure 2-10 (c)). It is also worth mentioning that if the two input

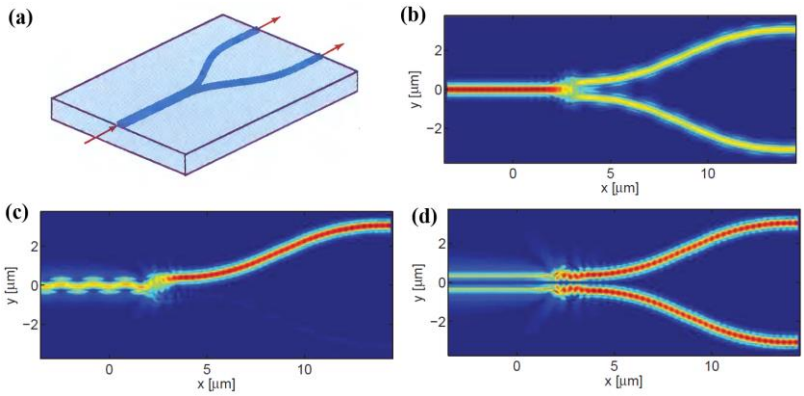


Figure 2-10: (a) Schematic presentation of Y-splitter (Saleh and Teich 2007), and electric mode profile in a Y-splitter in (b) splitting mode, (c) combing mode with only one input, and (d) combing mode with out of phase inputs (Chrostowski and Hochberg 2015).

ports in combiner modes are out of phase, destructive interference between the fundamental modes would result in zero output intensity (Figure 2-10 (d)) (Chrostowski and Hochberg 2015).

2.4 Interferometers

Interferometer is a general term for components in which a beam of light is split into two waves, and then combined, while one of them is delayed with respect to the other one. This delay causes a phase shift between the two interfering waves. Mach-Zehnder interferometers (MZI) are one of the simplest and yet most practical interferometers.

In an MZI structure, light is routed through two different arms using a 3dB coupler. The output is the result of interference between the waves from these two paths. An unbalance MZI has different path lengths. A simple representation is shown in Figure 2-11 (a). In such a structure the two waves experience the same propagation medium (same propagating constant, β) but the difference in propagation distances, Δd, creates

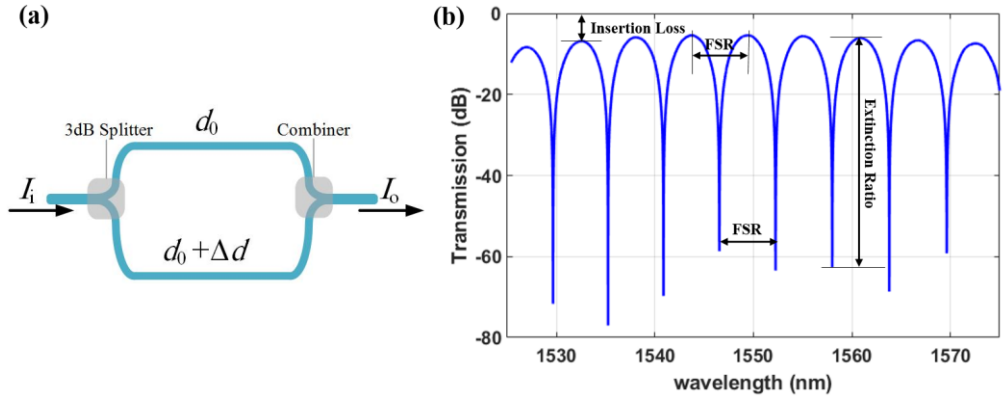


Figure 2-11: (a) Schematic presentation of unbalanced MZI interferometer, and (b) Transmission spectrum of a silicon nitride MZI with 100 μm length different.

a phase shift between the two interfering waves. The output intensity, I_o, as a function of input intensity, I_i, is given by:

$$I_o = \frac{1}{2}\left[1+\cos\left(\beta\Delta d\right)\right]I_i, \tag{2-3}$$

where β is defined in Eq. (2-1). Thus, the transmission of an unbalanced MZI is a function of the length difference Δd, which appears in the phase of the cosine function. According to this equation, when $\beta\Delta d$ is equal to $m\pi$ and m is an even integer number, $I_o = I_i$. However, for odd integer numbers, there is zero transmission at the output of MZI. The ideal transmission spectrum of MZI swings between zero and one as the wavelength changes. The period of this variation in the transfer function is known as free spectral range or FSR:

$$FSR = \Delta\lambda = \lambda_{m+1} - \lambda_m. \tag{2-4}$$

By assuming that the propagation constant changes linearly with respect to λ, the FSR is analytically obtained as

$$FSR = \frac{\lambda^2}{n_g \Delta L}, \tag{2-5}$$

where n_g is the group index of the waveguides. These equations suggest that, apart from the length difference, the FSR depends on the wavelength and the waveguide properties. A typical transmission spectrum of an interferometer is plotted in Figure 2-11 (b) which relates to an MZI with a 100 μm length difference. The waveguides have a silicon core and silicon dioxide cladding. The FSR for this MZI, as shown in the plot, is 5.8 nm.

Another important parameter in the performance of the MZI is the extinction ratio (ER). As shown in Figure 2-11 (b), the ER is defined as:

$$ER = 10\log\left(\frac{I_{\max}}{I_{\min}}\right), \tag{2-6}$$

In this sense, the ER is the measure of how distinguished the maximum and minimum levels are. Although acceptable values depend on the function and application of the device, reaching higher values is one of the design objectives.

The transfer function of an ideal interferometer oscillates between 0 and 1, but as it is illustrated in Figure 2-11 (b), it is not always the case. The loss in the different components, namely the couplers and waveguides, results in a deviation from 0 dB. This deviation is termed insertion loss and is analytically defined as:

$$\text{Insertion Loss} = 10\log\left(\frac{I_{\text{in}}}{I_{\min}}\right), \tag{2-7}$$

2.5 Tuning Mechanisms in PICs

On-chip tuning mechanisms reported in the literature today fall into three main subgroups. First is the thermo-optical effect, in which the refractive index is changed by modifying the waveguide temperature, (Saleh and Teich 2007). Silicon has a strong thermo-optical effect ($dn/dt = 1.86\times10^{-4}\ K^{-1}$ at 1550 nm measured in a relatively wide range of 5 to 300° K (Komma *et al*. 2012)) which makes it a suitable choice for thermo-optic tuning. Silicon nitride has a lower thermo-optical effect ($dn/dt = 4\times10^{-5}\ K^{-1}$ at 1550 nm (Amatya *et al.* 2007)) but it still can be effective for tuning the transmission spectrum of optical devices.

The thermo-optical effect, when used in integrated photonics systems, has a large tuning range (Pruessner *et al*. 2007). However, it suffers from limited operating speed

(response time in thermo-optic switches are on the order of microseconds to milliseconds) and high energy consumption (Du, Chau, and Zhou 2016).

Another common tuning mechanism is using electro-optical effects in doped silicon. Although smaller than its thermo-optics effect, doped silicon has a relatively good free-carrier plasma dispersion effect, which makes it possible to change its refractive index via changes in voltage. By changing the carrier (electron or holes) concentrations in a semiconductor, the absorption coefficient is also changed. In photonic systems, this is done by applying a reverse bias voltage to a heavily doped waveguide (Reed and Jason Png 2005). Using the plasma dispersion effect is also common in silicon modulators.

The response time of electro-optical switches is reported to be in the order of nanoseconds, and therefore, faster than thermo-optical mechanisms. A downside of this method is the change in optical absorption which occurs as the result of variation of carrier concentration in silicon. The outcome is an increase in insertion loss and crosstalk. Also, if not controlled, ohmic heating generated by the applied current induces a phase shift and increases the response time (Du, Chau, and Zhou 2016).

Both methods are based on the changes induced in the refractive index of the core waveguide and its cladding, and thus, have a limited range (less than 10% or in the range of 10^{-3} for silicon waveguide). This leads to a drastic increase in the size of the structure (on the order of cm) when more effect is desired (Chollet 2016). To overcome these limitations, a third mechanism is used which consists of micro or nano electro-mechanical system actuators. These mechanical actuators, which are known as microelectromechanical systems (MEMS) or nanoelectromechanical systems (NEMS), are not as fast but can induce greater changes and more flexibility by adding movable parts to the structure. MEMS actuators are relatively easy to integrate on-chip since

they are based on silicon material. This means they can be monolithically integrated with photonic devices in order to combine their fabrication process (Chollet 2016).

CHAPTER III

INTEGRATED OPTICAL FILTERS

Filters are now essential components in photonic integrated circuits. Their function is to select specific wavelengths transmitted on the waveguide, reshape signals, or optimize the signal to noise ratio in a link (Blumenthal *et al*. 2018). WDM filters are based on different devices such as arrayed waveguide gratings (AWGs), edge diffraction grating (EDGs), Mach-Zehnder interferometers (MZIs), waveguide Bragg gratings, and micro-ring resonators (MRRs) (Figure 3-1 (a) to (e)). What all these devices share, is their working principle which is interference, and therefore, their free spectral range (FSR) is determined by the interference order (Liu *et al*. 2021).

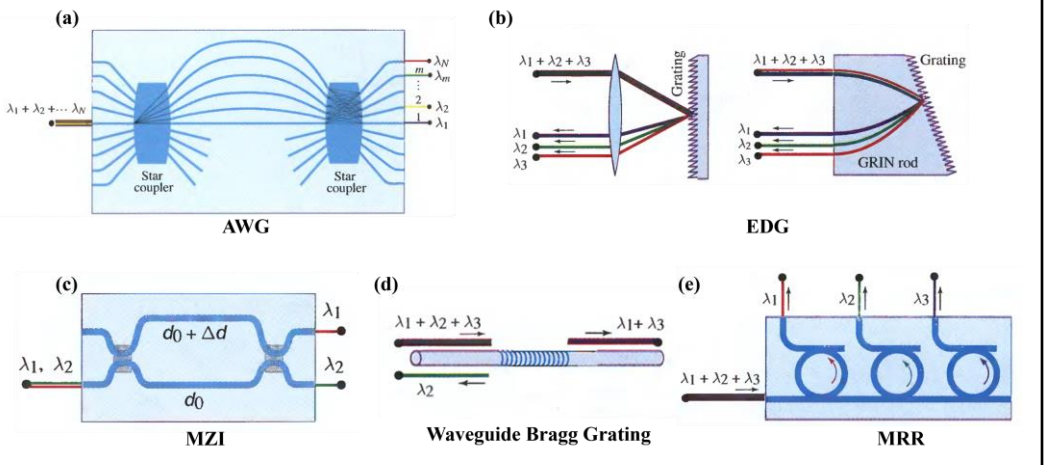


Figure 3-1: Schematics of different WDM filters (Saleh and Teich 2007)

The interference order in AWG filters is defined by the length difference between every two neighboring waveguides in the array. In an EDG the periodicity of the grating determines the interference order. These devices can offer 10 to 100 nm FSR enabling a large number of channels in a WDM system (Liu *et al*. 2021). In MRR filters, on the other hand, the interference order is limited by the ring radius. Silicon MMR can offer as low as 5 μm radius rings (Chrostowski and Hochberg 2015) and thus are more compact alternatives to AWGs. MZI based filters can provide larger bandwidth by using different coupling ratios and phase delay in cascaded MZIs. Cascaded MZIs are also used as inter-leavers for separating odd and even channels. The smallest interference order is obtained by Bragg gratings which is equal to 1.

In a WDM system, the desired filter design is the one that presents low loss, low crosstalk, and an acceptable 3-dB bandwidth with a flat-top response with sharp edges. Some designs also require insensitivity to polarization and thermal changes (Liu *et al*. 2021).

Early filter designs were based on waveguide gratings (Trinh *et al*. 1997; Suzuki *et al. 1997)* but now more research is done on Fabry-Perot cavities and micro-ring or disk resonators. Bragg gratings are still used in wavelength division multiplexing applications, but they are very sensitive to fabrication variations and therefore hard to implement. Besides, multichannel filtering is challenging using a waveguide grating since adding more gratings to the device increase the insertion loss (Saber *et al*. 2018).

Fabry-Perot cavity filters are standing wave resonators in which the comb-like output result of the filter is obtained by multiple interferences that take place inside the Fabry-Perot interferometer (Saleh and Teich 2007). A basic Fabry-Perot (FP) cavity consists of two mirrors separated by a distance of L. The light that enters the cavity

will go through multiple reflections between the mirrors and the interference between these reflections build up the cavity modes. Therefore, the interference determines the shape of the output. In the integrated systems, reflections inside the cavity can be obtained with grating couplers or Sagnac loop mirrors (SLM).

3.1 Proposed Designs: Fabry-Perot Filter Using Sagnac Loop Mirrors

Here, we will focus on devices using multiple interferences in cavities, such as Fabry-Perot filters. The basic building blocks of such integrated optical cavities are a waveguide and two reflecting structures or mirrors. Often, these cavities are formed by patterning partially reflecting Bragg mirrors along an input waveguide, but it is challenging to build this type of mirror. Therefore, we will investigate configurations without partial Bragg mirrors or where the input signal does not have to be transmitted through a grating. Hence, we aimed to obtain fabrication tolerant devices with minimal insertion losses.

3.2 Mach-Zehnder Assisted Sagnac Loop Mirror Cavity Design

The first version of this filter is designed using Sagnac loop mirrors (SLMs) and multi-mode interference (MMI) couplers. A MZI is added before each reflector to create the phase difference required to control their reflectivity. The FSR of this device is determined by the overall length of the cavity, which corresponds to the propagation distance of the waves through the MZI arms, SLMs, and the waveguide connecting the two reflectors. To obtain large FSRs, the overall length of the cavity should be short. Although SLMs are considered robust and wavelength independent structures, their large bending radius limits this design. Here we are using SiN waveguides with a

thickness of 440 nm and widths of 450 or 850 nm. When using such waveguides, the bending radius should be greater than 100 µm to ensure minimal losses. This restriction should be considered for all bended or curved waveguides in the device.

This relatively high bending radius also affects the MZI design. However, with a change in configuration of the MZI, we designed, optimized, and fabricated a compact MZI which provides the required phase difference. In this design, instead of two arms with different lengths, the arms had the same length but different widths (resulting in different effective refractive indices). The schematic of the filter using this MZI is presented in Figure 3-2.

Nevertheless, it was not possible to obtain a 400 GHz FSR, large enough to enable the isolation of a single channel in a NG-PON2 receiver, the target application of our industrial collaborator in this project. Thus, SLMs implemented with SiN waveguides are not suitable for this application.

3.3 Bragg Grating Reflector Cavity Design

Considering these constraints and requirements, an alternative reflection mechanism must be used to replace the SLMs. Thus, in the next part of this project, the

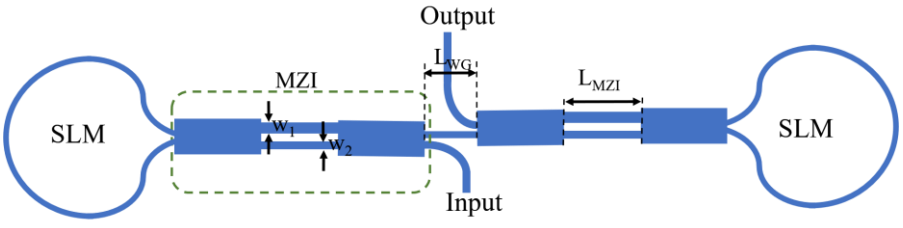


Figure 3-2: Schematic of a Fabry-Perot Filter using Sagnac loop mirrors and compact MZI

focus will be on the design of broadband integrated Bragg reflectors to replace the SLMs. As shown in Figure 3-3, the signal will never be transmitted through the reflectors. Hence, we will be able to use short gratings with a strong refractive index modulation that are reflective over a broad wavelength range. Although small variations of the grating dimensions due to fabrication errors would shift spectrum, the broad wavelength range guarantees that the reflectance will cover the desired band.

Also, MZIs are no longer needed since the required phase difference can be obtained by changing the length of the waveguide connecting the grating reflectors to the MMI coupler. The schematic of this filter design is shown in Figure 3-3. As in the first design, the waveguides before the gratings contribute to the overall length of the cavity and will be used to tune the FSR of the filter. Thus, replacing the SLMs, not only increases the FSRs but can also reduce the overall size of the device resulting in a more compact filter structure.

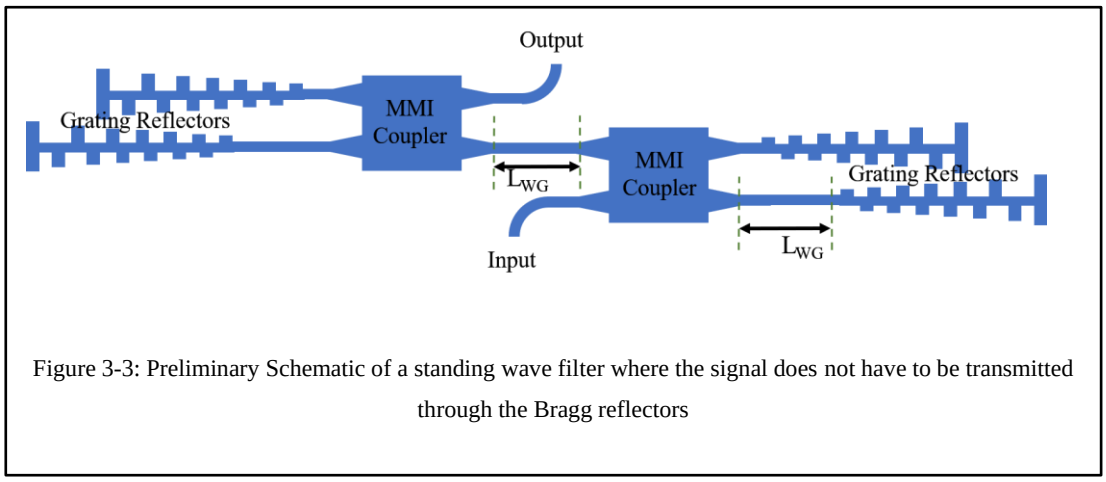
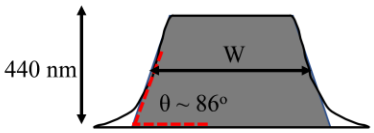


Figure 3-3: Preliminary Schematic of a standing wave filter where the signal does not have to be transmitted through the Bragg reflectors

(a)



(b)

W = 850 nm	n_{eff}	n_g
Fabrication	1.6455	2.1348
Simulations	1.6472	2.1325

Figure 3-4: (a) Schematic of the fabricated waveguide shape and the simulated model. (b) a table showing the measured effective and group index of the waveguide compared to the values obtained from simulations.

3.4 Fabrication and Design Challenges

The proposed structure was fabricated using 3.2 µm of silicon dioxide (SiO_2) deposited on a silicon wafer as the bottom cladding and a 3.4 µm layer of SiO_2 as the top cladding. Both cladding layers were made with low pressure chemical deposition (LPCVD) of tetraethoxysilane (TEOS), and the core layer is LPCVD SiN. Surface grating couplers were used to couple the light in and out of the chip.

The waveguides are patterned using photolithography. Through common exposure and etching processes, a perfectly vertical etch on the sidewalls of the waveguides is not possible to achieve. Figure 3-4 (a) presents the shape of the waveguide after the etching process. The width shown by W is the waveguide width which is the nominal value targeted by the photolithography. The black outline in Figure 3-4 (a) is the actual shape of a fabricated waveguide. However, in simulations, this shape is modeled by the gray trapezoid. In the latest fabrication done by our industrial partner AEPONYX at the C2MI, SiN waveguides with a thickness of 440 nm and a width of 850 nm have a 86° sidewall angle. The measured effective and group index of these waveguides are shown in Figure 3-4 (b).

In a photonic circuit, in most cases, bended waveguides are unavoidable. Similar to fiber optics, having bent waveguides in PICs can add unwanted losses to the system. The loss in a bent waveguide consists of mode mismatch losses and radiation losses. The mode mismatch or junction losses arise from the difference in the modes supported by the straight waveguide and the bended waveguide. Radiation losses can be compensated by increasing the width of the waveguide since it increases the modal confinement.

To measure the bending losses in a waveguide, the cut back method is used. In this method, waveguides with a different number of bends similar to Figure 3-5 (a) are fabricated and their spectrum is measured and plotted in Figure 3-5 (b). The waveguides have 8, 24, 48, and 96 bends of 90 degrees each. Thus, as the number of

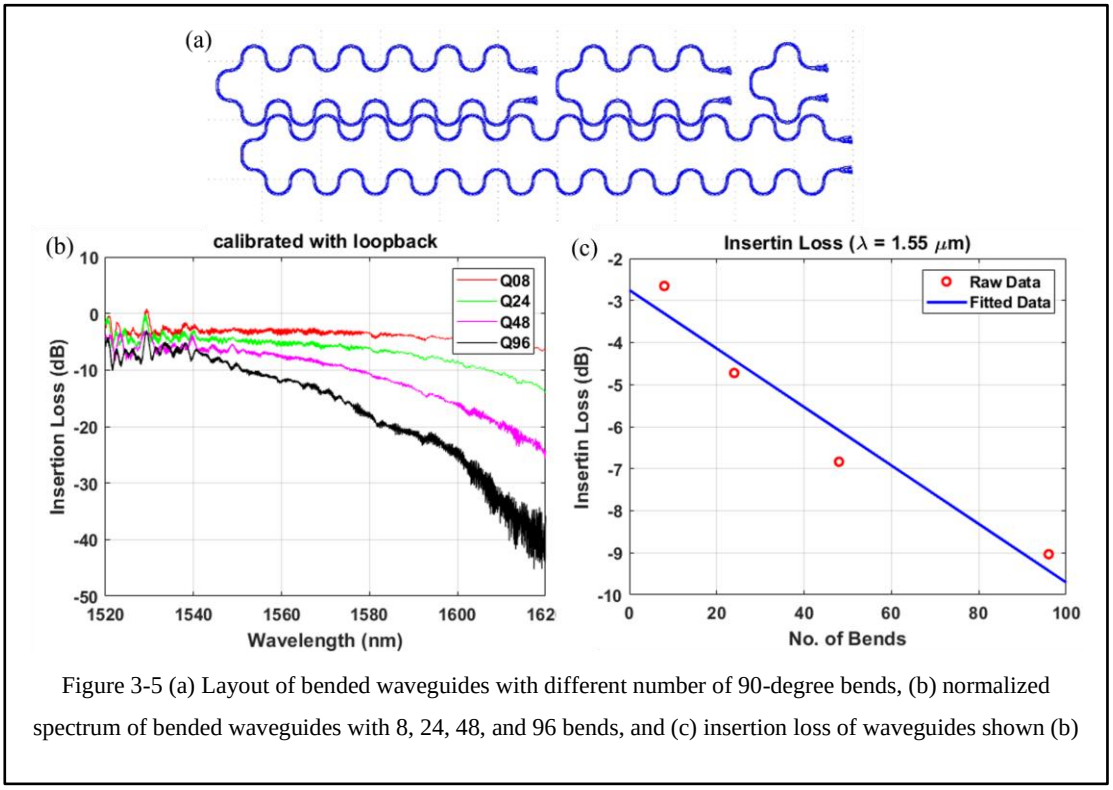


Figure 3-5 (a) Layout of bended waveguides with different number of 90-degree bends, (b) normalized spectrum of bended waveguides with 8, 24, 48, and 96 bends, and (c) insertion loss of waveguides shown (b)

bends increases, the corresponding losses will also increase. This loss consists of mode mismatch, radiation, and propagation losses. The total loss of each 90 deg bend can be calculated by the cut-back method and the plot in Figure 3-5 (c). This plot shows the regression of the insertion loss of the four bended waveguides versus the number of the bends. The total bending loss (including propagation losses) for the corresponding radius of curvature at the specific wavelength is the slope of this plot.

The waveguides used in this experiment were SiN waveguides with a thickness of 435 nm and width of 450 nm and a radius of curvature is 100 μm. The measurements show a total loss of 0.21 dB per bend for these waveguides. As mentioned before increasing the mode confinement by increasing the width of the waveguide will help in decreasing the bending loss. Thus, in the following designs and fabrication, a waveguide width of 850 nm is used.

CHAPTER IV

COUPLERS: DESIGN AND ANALYSIS

Power splitter or couplers are an essential part of optical filters. All three proposed designs here use multimode interference (MMI) couplers as 3 dB power splitters due to their fabrication tolerance and compactness. Thus, as the first step in this research, we start with the analytical study, design, and optimization of 1×2 and 2×2 MMI couplers.

4.1 Brief Review

As previously discussed in chapter II, the operating principle of MMI couplers is based on imaging theory and interference. There are three different interference regimes to build up the image in the multimode region. The first is general interference which is independent of the location of the input port. The other two are the paired and symmetric interference regimes, which highly depend on the position of the input since they are based on the selective excitation of modes. Table 4-1 summarizes the requirements and characteristics of these three regimes (Soldano and Pennings 1995).

Table 4-1: Characteristics and requirements of different interfering regimes (Soldano and Pennings 1995)

Interference regime	General	Paired	Symmetric
Input × Output	N×N	2×N	1×N
First N fold image distance	$3L_\pi/N$	L_π/N	$3L_\pi/4N$
Input location	any	$\pm W_e/6$	0

In this table, L_π is called the beating length of the coupler and is calculated based on the propagation constant of the fundamental and first order mode in the multimode region of the coupler:

$$L_\pi = \frac{\pi}{\beta_0 - \beta_1} \tag{4-1}$$

Also, W_e is slightly larger than the multimode region width since it also accounts for the penetration depth in the cladding of the modes. Nevertheless, in structures with high contrast between the core and the cladding, W_e can be considered equal to the width of the multimode waveguide.

4.2 Design and Simulations

The eigenmode expansion (EME) method in Lumerical software was used to simulate and design the MMI couplers. A 1×2 MMI coupler always work in the symmetric interference regime. If the locations of the output ports are optimized, in simulations the power of both output ports are equal. The design and optimization of a 2×2 MMI is more challenging.

Figure 4-1 shows the schematic, mode profile, and output power of one of the designed 2×2 MMI coupler. Figure 4-1 (b) clearly shows the formation of the images at the output ports. The tapers help the mode to evolve slowly and reduce the coupling losses. However, Figure 4-1 (b) shows that still a part of the power is lost at the tapers due to scattering. A longer taper can compensate for this loss but would also increase propagation losses.

We monitored the S-parameters at the output ports to study both the insertion loss

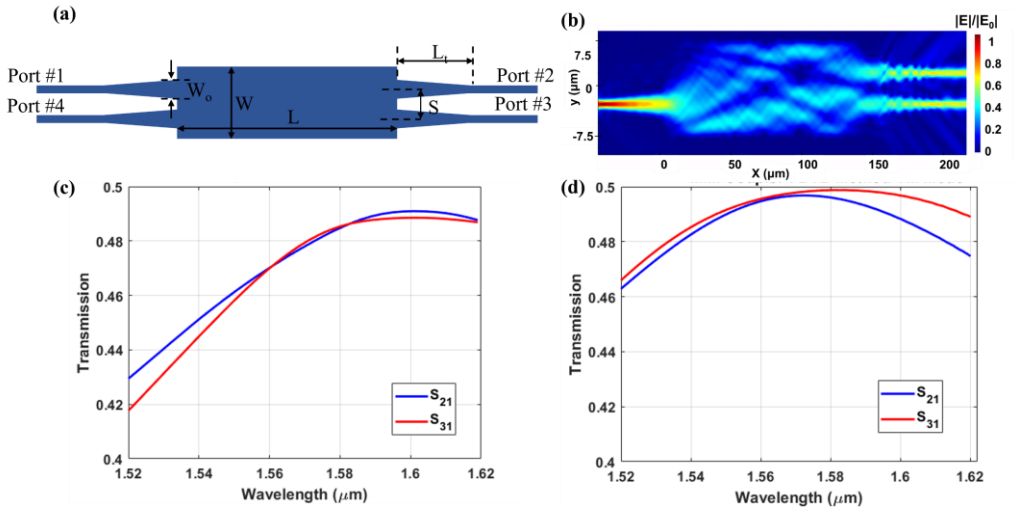


Figure 4-1: (a) Schematic of a 2×2 MMI coupler, (b) distribution of the normalized electric filed in the coupler for the TE mode, and output transmission power for ports 2 and 3 in (c) TE and (d) TM modes.

and coupling ratio. The dimensions of the coupler are optimized for the L band and as Figure 4-1 (c) shows, we have achieved a good coupling ratio at 1597 nm. At this wavelength, the coupling ratio for ports 2 and 3 are 0.503 and 0.496, respectively. The calculated insertion loss is -0.124 dB.

Although the MMI couplers were all designed for the TE mode, the same structure has been tested with the TM mode to investigate its polarization sensitivity. As Figure 4-1 (d) shows, the TM spectrum of the device is flatter in the wavelength range of 1520 to 1620 nm, but the lowest insertion loss is not at the center of the L band anymore.

4.3 Fabrication and Measurement

We optimized and prepared multiple 1×2 and 2×2 MMI couplers. Some of these structures were fabricated and measured by AEPONYX Inc. In these sets of

fabrications, the coupling and out coupling of light to the PICs are done with surface grating couplers. These couplers can sometimes be very lossy and usually do not have a flat spectrum. Thus, to remove the effects of the surface grating on the response of the MMI couplers, a loopback structure consisting of only a short waveguide connecting two surface gratings is added to the dies close to the devices. The spectrum of the loopback is then measured and used to normalize the output response.

Figure 4-2 shows an example of the loopback spectrum and the output response of a MMI. These plots show that the surface gratings that we used are noisy at shorter wavelengths and that their best insertion loss is -33.41 dB (about 16.7 dB per surface grating coupler). This also affects the measured spectrum of the MMI couplers. As the plots in Figure 4-2 (b) show, the response cannot go higher than -37 dB, is very noisy at shorter wavelengths, and has the dome shape of the surface grating spectrum.

The measurement results and calculations for a 1×2 MMI are presented in Figure 4-3. The width and length of the MMI region of this coupler are 8 and 37.4 μm, respectively. The separation between the outputs is 4.10 μm and the width of the taper

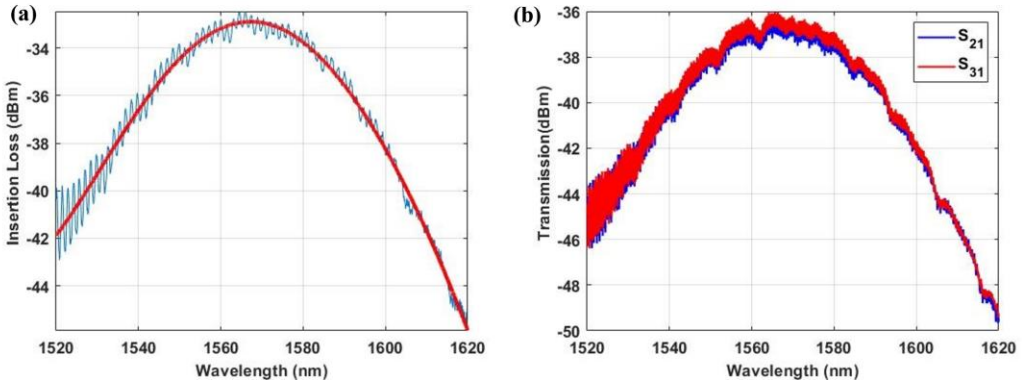


Figure 4-2: Measured spectrum of the (a) loopback structure and (b) a 1×2 MMI.

at the output is 2.87 μm. We have separately optimized the width of the taper at the MMI region since a small width would not cover the spot size of the formed images and thus would not collect maximum power. A width that is too large, on the other hand, would result in the coupling of unwanted modes in the output waveguides, which in turn affects the splitting ratio.

Figure 4-3 (a) and (b) show the spectrum before and after normalization, respectively. One can see that the noises from the grating couplers make the calculations difficult. One solution to mitigate this noise is to use an average of the spectrums instead. A fitted polynomial of the third degree can effectively model the

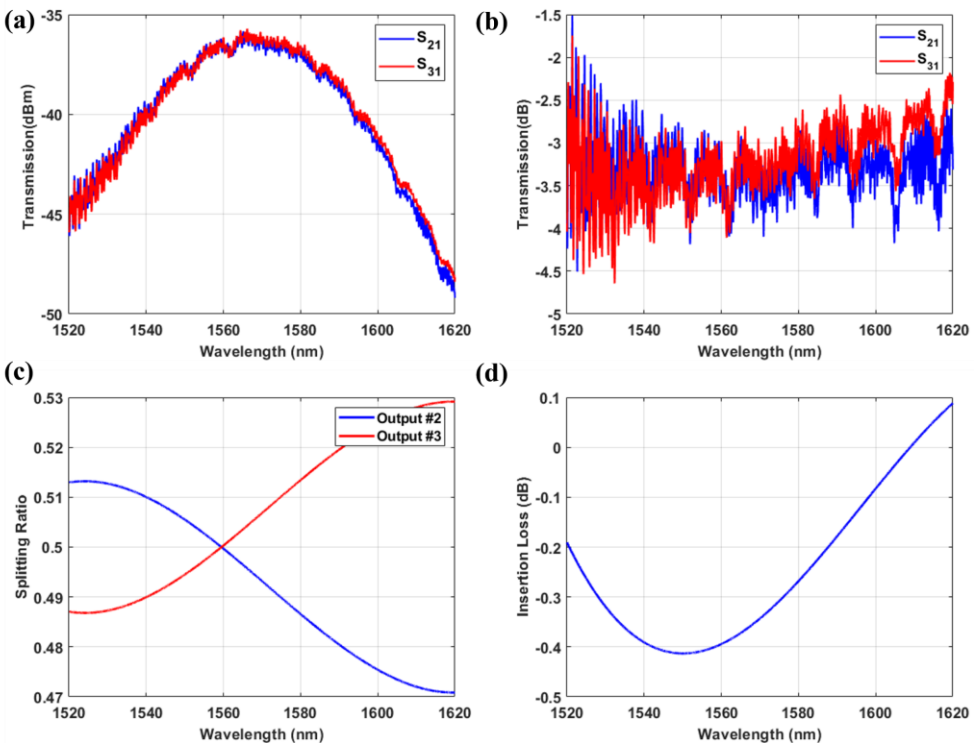


Figure 4-3: (a) Raw spectrum of the two outputs of MMI, (b) normalized spectrum, (c) splitting ratio, and (d) insertion loss of the MMI couplers.

output spectrums. Using these fitted curves, we have calculated the splitting ratio and the insertion loss which are presented in Figure 4-3 (c) and (d). As expected, the coupler has a low insertion loss in the L band. However, the plot in Figure 4-3 (d) shows that the insertion loss keeps decreasing even beyond zero dB which is an error due to the noises in the spectrum.

The same results are presented for a 2×2 MMI coupler in Figure 4-4. The coupler has a width and length of 10 and 76.3 μm, respectively. The separation between the outputs is 3.72 and the width of the taper at the output is 2.87 μm.

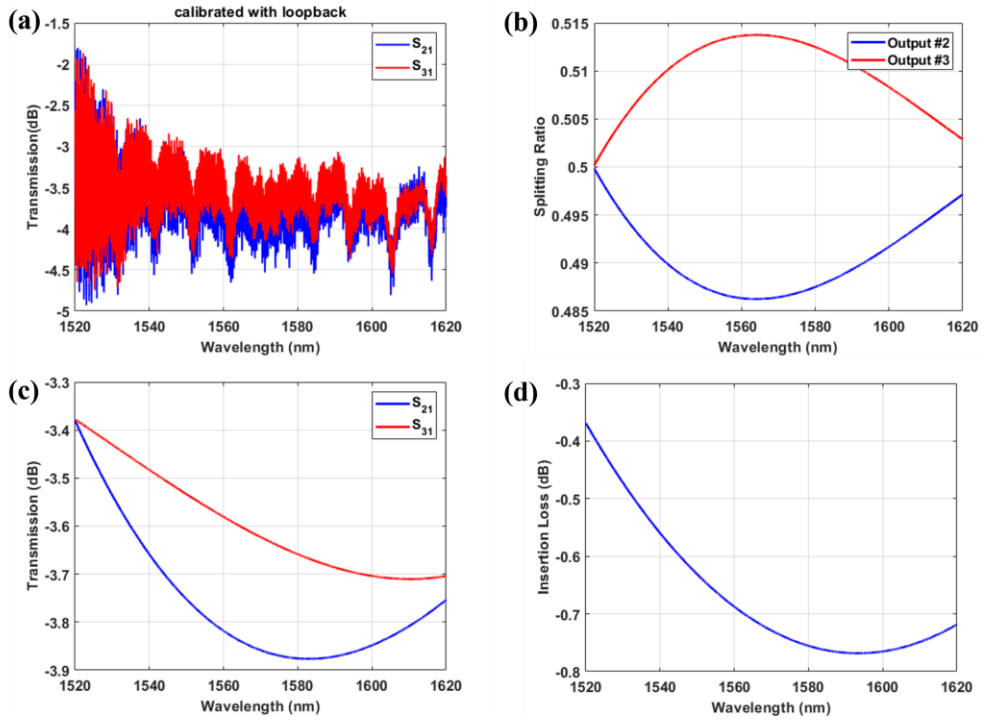


Figure 4-4: (a) Normalized spectrum of the two outputs of MMI, (b) Splitting ratio, (c) Transmission, and (d) insertion loss of the 2×2 MMI couplers.

Although the insertion loss of this coupler is larger than for the 1×2 MMI coupler, the splitting ratio is slightly better in the L band. We also show the transmission in Figure 4-4(c), which is a fitted plot of the output spectrum. The insertion loss plotted in Figure 4-4 (d) follows the same trend as the transmission spectrum.

In this chapter, we presented the design, simulations, and fabrication results of 1×2 and 2×2 MMI couplers. Although we only need 2×2 MMI couplers for our filter design, we investigated 1×2 MMI couplers as 3-dB power splitters. The MMI coupler presented in Figure 4-4 is the best design according to the fabrication results and thus we are going to use this coupler in the devices of the following chapters. The measurement results from six different dies confirm a minor mismatch in the splitting ratios of the couplers. The average of this mismatch between six dies is 1.38%. These results confirm the fabrication tolerance characteristics of MMI couplers in comparison to similar structures.

CHAPTER V

COMPACT MACH- ZEHNDER INTERFEROMETERS

Since the refractive index contrast between SiN and SiO_2 is lower than in silicon photonic devices, this creates challenges specific to this platform. Unlike silicon photonics, waveguides on SiN platforms cannot have sharp bends. The fundamental mode is less confined in the waveguide core and thus a SiN bent waveguide with a small radius radiates a significant portion of the power. To avoid bending losses, SiN waveguides are limited to large radiuses of curvatures (Han, Ménard, and Shi 2020), which in turn results in devices with a large footprint. This reduces the integration density achievable in SiN platforms in comparison to silicon photonics, which can impact the size and cost of optical systems. Different approaches have been demonstrated to further miniaturize integrated optical devices, including the use of plasmonics (Cheben *et al*. 2018), creating multifunctional devices (Shimogaki *et al*. 2021) that can replace two components, or re-arranging the devices on the chip to minimize the space between them (García-Meca *et al*. 2017). Nevertheless, each of these solutions has important drawbacks such as higher propagation losses, limited efficiency, and higher crosstalk.

In this chapter, we demonstrate an unbalanced Mach-Zehnder interferometer (MZI) implemented on a SiN platform that avoids the trade-off between size and bending losses by creating an effective index difference between the two arms instead of using

different lengths to create the phase difference. This modification achieves a reduction in total device footprint up to 50 times in comparison with the traditional approach.

5.1 Theoretical Analysis and Design

The phase accumulated by an optical wave propagating through a waveguide is defined as $e^{-j\beta l}$, where $j = \sqrt{-1}$ is the imaginary number, l is the propagation length and β is the propagation constant of the mode. At a specific wavelength, λ, the propagation constant is defined as $\beta=(2\pi n_{eff})/\lambda$. In this relation, n_{eff} represents the effective refractive index of the propagating mode which depends on the waveguide shape, dimensions, and material.

In a conventional unbalanced MZI, a phase shift is obtained by creating a length difference between the two arms of the interferometer. Therefore, the phase difference is equal to $\beta\Delta l$ since both waveguides have the same propagation constant. Alternatively, if the propagation constant is altered instead of the propagation length, the phase difference will be given by $\Delta\beta l$, where $\Delta\beta$ is the difference between the propagation constant in each arm. For SiN waveguides with a fixed core thickness, changing the width of the waveguide directly affects the effective refractive index of the mode, which in turn changes the propagation constant. Therefore, by employing waveguides of the same length but with different widths as the MZI arms, one can still generate a phase shift at the outputs.

As the schematic in Figure 5-1 (a) shows, this MZI configuration does not require any bends, which reduces the footprint of the device and avoids bending losses (mode mismatch and radiation losses). Considering that a bending radius of about 100 µm is required to minimize bending losses in SiN waveguides with a width of 850 nm and thickness of 440 nm (Han, Ménard, and Shi 2020), the footprint of the proposed MZI is reduced by a factor of 46 in comparison to a conventional MZI with the same free spectral range (FSR). This is mainly due to the fact that, as shown in Figure 5-1 (b), at least four 90 degrees bends are required to create the length difference Δl. As a result, the couplers forming the MZI must be at least 400 µm apart and the minimum optical propagation length is 628 µm. It is possible to reduce the minimum number of bends

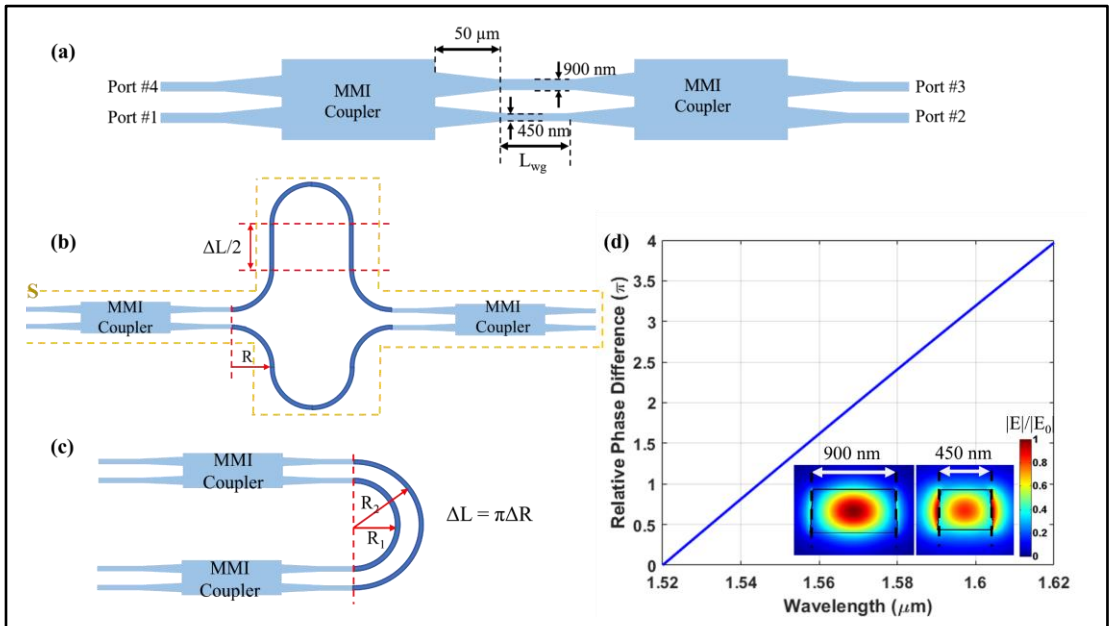


Figure 5-1: (a) Schematic of the proposed MZI structure and dimensions of its components, (b) Schematic of a conventional MZI with a bending radius of R, (c) schematic of a folded MZI design to reduce the footprint of conventional MZIs, and (d) calculated relative phase difference between the two arms for L_{wg} = 150 µm as a function of wavelength. The inset shows the normalized electric field intensity of the fundamental TE mode across the waveguides of the MZI presented in (a).

needed to two by folding the MZI, as shown in Figure 5-1 (c). Also, this architecture imposes a minimum length difference between the arms equal to the separation between the waveguides at the output of the couplers multiplied by 2π (Armin, Nabki, and Ménard 2020). Nevertheless, the configuration with different waveguides is still at least 70 times smaller.

Furthermore, to have a robust device, multimode interferometer (MMI) couplers are used instead of directional couplers. Although these couplers have a slightly higher insertion loss than directional couplers, they provide a more accurate splitting ratio and have a lower sensitivity to fabrication variations. The multimode interference region for the coupler used in this design is 76 by 10 μm and is connected to the waveguides with a 50 μm long taper. This is one of the couplers that has been designed, optimized and fabricated in prior to the filter design. The design and the measurement results are presented in chapter 4. The wide side of the tapers is 3.2 μm and their width decreases to match the waveguide width at the other end, alloying for the modes to evolve gradually to avoid losses. The input and output waveguides of the MZI have a width of 850 nm.

The width of the upper and the lower MZI arms in this design are 450 and 900 nm, respectively. The core layer is 440 nm thick, and thus both waveguides are single mode in the optical C and L bands. The 900 nm wide waveguide is at the limit of single mode operation whereas the 450 nm one is the narrowest can be reliably fabricated with our process. Therefore, these widths provide the largest $\Delta\beta$ that we can implement while remaining single mode. The distribution of the electric field intensity along the cross-section of the two waveguides is illustrated in Figure 5-1 (d) as insets showing the

distinctive confinements of the fundamental transverse electric (TE) modes in these two geometrically different waveguides.

The tapers connecting the MMIs to the waveguides forming the MZI arms end with different widths and thus have a different rate of change. This results in a phase difference between the optical waves propagating in the tapers, even before the mode reaches the waveguides. Therefore, the propagation along the length of the tapers must also be considered in the analytical calculations of the phase difference. To find the phase change accumulated in the tapers, we first calculated the effective index of the fundamental TE mode at intervals of 2.5 μm along the tapers. Then, through interpolation, we obtained equations for the effective index change along the tapers and used them to calculate the accumulated phase as a function of the length of each taper. The results of these calculations are shown in Figure 5-1 (d) where the plot indicates that for a MZI with 80 μm long waveguides and tapers having a length of 50 μm, a 2π phase shift can be reached in the L band.

5.2 Results and Discussions

The proposed structure was fabricated using 3.2 μm of silicon dioxide (SiO_2) deposited on a silicon wafer as the bottom cladding and a 3.4 μm layer of SiO_2 as the top cladding. Both cladding layers were made with low pressure chemical deposition (LPCVD) of tetraethoxysilane (TEOS) and the core layer is LPCVD SiN. Surface grating couplers were used to couple the light in and out of the chip. An image of the fabricated device is presented in Figure 5-2 (a). Ports 2 and 3 in this figure are the outputs that correspond to S_{21} and S_{31}, respectively. Figure 5-2 (b) presents the

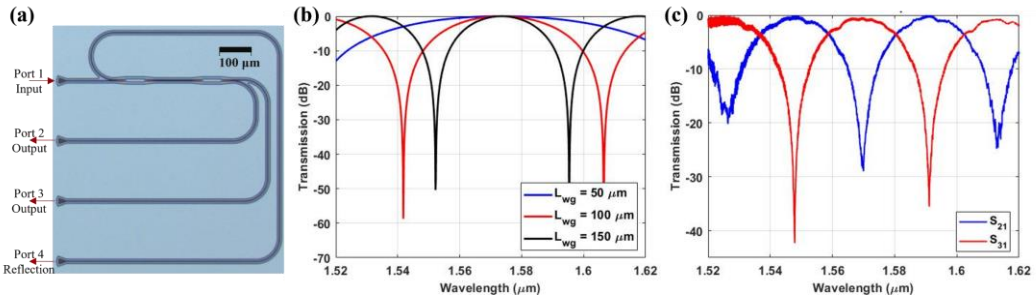


Figure 5-2: (a) Image of the fabricated device, (b) simulated results for the transmitted power at one output of the MZI for three different waveguide lengths, (c) measurement results of the transmitted power at the outputs of a fabricated MZI with an arm length of 150 µm.

simulation results of the transmitted power at the outputs of the novel MZI structure for three different lengths of the MZI arms, L_{wg}. The simulations were carried out using the eigenmode expansion method in the MODE software from Ansys Lumerical. As expected, an increase in L_{wg} results in a decrease in the FSR of the interferometer. Increasing the propagation length in the MZI arms leads to a larger phase difference at the outputs. This is equivalent to increasing the length difference in a conventional MZI.

The plot presented in Figure 5-2 (c) shows the measured data after normalization with a reference structure to cancel out the effect of the grating couplers. The results correspond to L_{wg} = 150 µm and are in good agreement with the simulated ones. The average insertion loss of the device is 0.42 dB, which was measured using six different dies from three wafers. As observed in all the dies, the extinction ratio at each output of the MZIs are not equal. This is caused by a slight deviation in the splitting ratio of the MMI couplers. Standalone MMI couplers were fabricated on the same dies as the MZIs. The measurement results confirm a minor mismatch in the splitting ratios of the couplers. The average of this mismatch between six measured dies is 1.38%. These results also revealed that, as expected, the main source of loss in the MZI is the insertion loss of the couplers.

Both the proposed design and the conventional MZI require changes in the length of the waveguides forming the arms of the interferometer to tune the FSR. However, in the proposed design the lengths of both arms must change equally to achieve the phase difference required for a given FSR, whereas in the conventional design only one of the two arms should be altered. Figure 5-3 compares the change in the FSR as the lengths of the waveguide arms are changing for both types of MZI. In the first type, L_{wg} is the total length of the waveguides whereas, in the conventional MZI, ΔL is the length difference between the two arms. Thus, $\Delta L = 0$ means that there is no phase difference between the arms (i.e. the MZI is balanced) resulting in an infinite FSR. However, as explained above the minimal waveguide length for the arms of a conventional MZI must be at least equal to four 90 degrees bends, in addition to the value of ΔL.

Although input and output waveguides of the MZI have a width of 850 nm, the narrow and wide waveguides of the interferometer have widths of 450 and 900 nm,

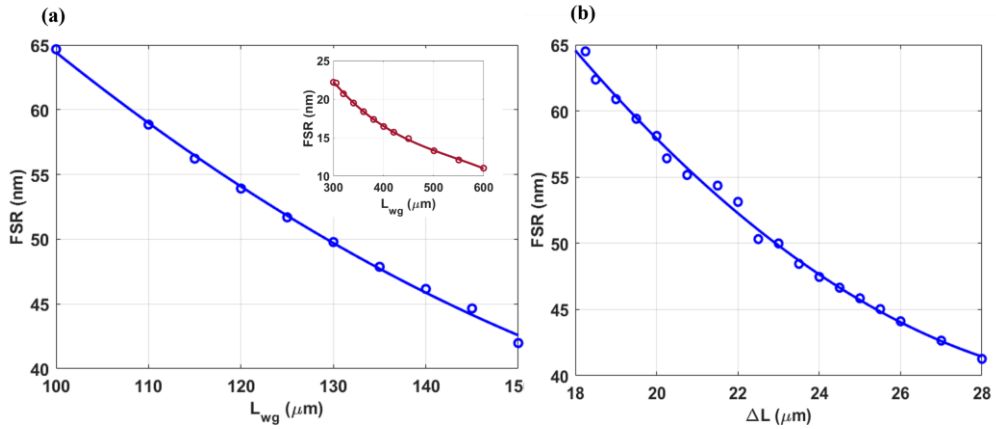


Figure 5-3: (a) Total MZI arms lengths required to obtain a FSR between 65 and 42 nm with the proposed design compared to (b) the length difference needed in a conventional MZI for the same FSR. The inset in (a) shows the corresponding FSRs for longer length of the MZI arms in the proposed design.

respectively. Thus, the phase difference $\Delta\varphi$, in this case, results from a fixed propagation constant difference multiplied by the length of the waveguides, i.e., $\Delta\varphi = \Delta\beta L_{wg}$. As we limit ourselves to single-mode SiN waveguides that are 440 nm thick, the difference between the effective indices is always less than 0.17, which means that $\Delta\beta$ always stays below 6.5×10^5 rad/m in the L band. On the other hand, the phase difference in a conventional MZI is given by $\Delta\varphi = \beta\Delta L$, where the propagation constant has values of more than 5.5×10^6. This explains why larger changes in waveguide lengths are needed in the proposed design to modify the FSR when compared to conventional MZIs. Nevertheless, even with the long waveguide lengths required to maintain the FSRs below 10 nm, the proposed design is still more compact than conventional MZIs. It does not matter how small the length difference in a conventional MZI is, large bending radiuses are still needed to limit losses in curved SiN waveguides, which leads to a significant device footprint. To illustrate this, one can have an FSR of 42.83 nm with the proposed design if the waveguides forming the arms of the MZI are 150 μm long. The same FSR can be obtained with a 27.2 μm length difference (ΔL) in a conventional MZI with 850 nm wide waveguide arms. Nevertheless, by looking at the schematics shown in Figure 5-1 (a) and (b), a conventional MZI using SiN waveguides will occupy more than 40 times the floor space of the proposed design due to the large bending radiuses required to implement it.

5.3 Tunability

Sensitivity to fabrication variations is a challenge in photonic integrated circuits. Using robust components, such as MMI couplers instead of directional couplers, can help to mitigate this. However, the interference in this design is a function of the effective refractive indices of the waveguides, which are in turn dependent on the

physical dimensions and thus are affected by fabrication variations. Moreover, the fact that the width of the waveguides used as the MZI arms are not the same makes it more sensitive to fabrication variations. In a conventional MZI, where all the waveguides have the same width, a slight change in the dimensions of the device modifies the propagation constant in each of the MZI arms equally. This has a limited effect on the phase difference between the arms. Conversely, in the proposed MZI a variation in the dimensions of the arms changes the propagation constant in each of them differently, which can significantly affect the phase difference accumulated through the device. Variations caused by the fabrication process can be compensated by implementing a tuning mechanism. Although the thermo-optical coefficient of SiN is lower than the one of Si, thermo-optical effects can be used to fine tune the response of the MZI. Fortunately, as will be demonstrated below, having waveguides with different dimensions can simplify the design of thermal actuators.

For this study, we consider a simple and easy to fabricate 15 μm wide aluminum (Al) wire, located on top of the MZI arms as a local heater. The changes in the effective index due to thermo-optical effects in the waveguides were calculated with the Finite-Difference Eigenmode (FDE) method and are presented in Figure 5-4. These plots also show the corresponding change in accumulated phase for 100 μm long waveguides caused by the variation in temperature. The temperature is controlled by changing the power dissipated by the heater, and thus, the results are plotted as a function of power.

Since the MZI is very compact, its arms are separated by less than 5 μm, which corresponds to the separation between the outputs of the MMIs. Considering a 15 μm wide Al wire, both waveguides will experience the same temperature change as the

power dissipated by the heater increases. Yet as shown in Figure 5-4, the change in the accumulated phase as a function of power dissipated by the heater is significantly higher in the 900 nm wide waveguide in comparison to the phase change occurring in the 450 nm wide waveguide. Therefore, it is possible to effectively tune the response of this compact MZI with a single heater. Figure 5-4 (d) indicates that a 5.47 nm shift is expected in the output spectrum of the MZI when the power of the Al heater is increased to 100 mW.

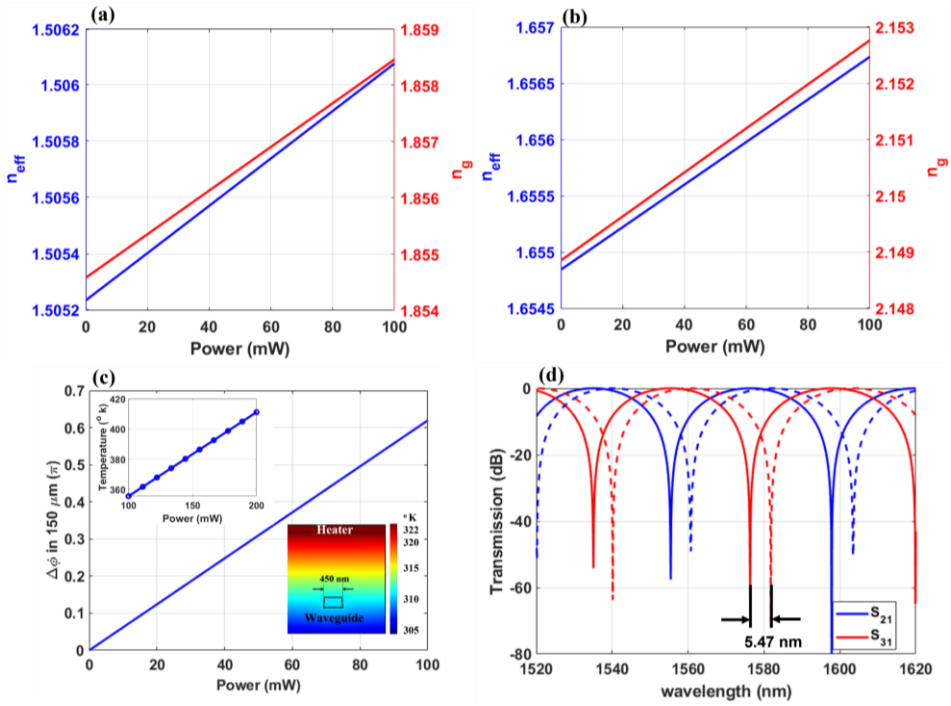


Figure 5-4: Modeling of thermal effects on the effective index (shown in blue) and group index (shown in red) of a waveguide with a width of (a) 450 nm, and (b) 900 nm. (c) Change in the accumulated phase between the narrow and wide waveguides after propagation over a length of 100 µm as a function of power dissipated by the heater. The insets show the temperature at the center of the waveguides versus heater power and the heat map around a 450 nm waveguide when P = 100 mW. (d) Spectrum of the MZI when the Al thermal heater dissipate a power of 100 mW. The dashed lines show the spectrum of the MZI when the local heater is not working.

Figure 5-5 shows an example of compensation of fabrication variations with a local heater. A 10 nm increase in the width of the MZI arms results in a shift of 7.33 nm of its spectrum, as shown in Figure 5-5. This can be corrected by increasing the temperature of the MZI arms by 67.86^{o} C, which corresponds to a power dissipation of 118 mW.

The compact design does have limitations with regards to tuning power efficiency in comparison to conventional MZIs. The large bend waveguides required in conventional MZIs make it possible to thermally isolate each arm. Thus, a large temperature gradient can be created between the arms by heating only one. As a result, the phase difference as a function of dissipated power is larger in conventional MZIs. For instance, to obtain the same FSR as the one shown in Figure 5-5 with a conventional MZI built with waveguides that are 850 nm wide, the length difference between the arms must be 27.2 μm. By heating only one of the arms, a shift of 11.5 nm can be obtained with 100 mW of dissipated power. Therefore, for the same amount of power, the conventional MZI provides a 57% larger spectral shift over the compact MZI configuration.

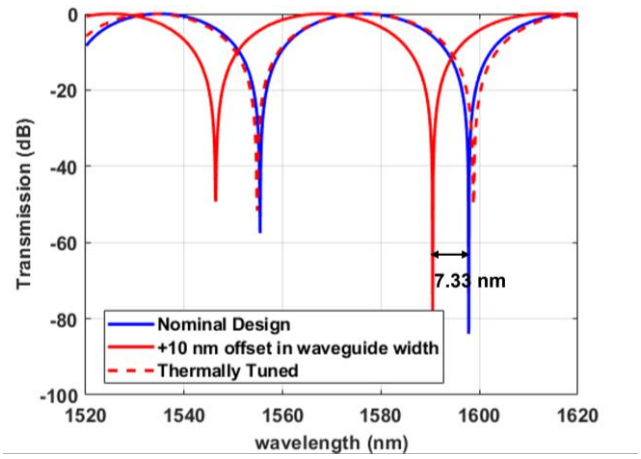


Figure 5-5: Changes in the output of the one of the MZI ports imposed by 10 nm deviation in the width of the waveguides and the results of the thermally tuning

In this chapter, we proposed and demonstrated a compact MZI for SiN platforms that was experimentally characterized in the C and L bands. We also showed that by adding a simple local heater it is possible to overcome the fabrication sensitivity. We can take advantage of the different changes in effective refractive index between the narrow and wide waveguide to tune the response of the MZI, with a single simple and compact heater. The experimental measurements are in agreement with simulations and analytical calculations. The reduced footprint of the proposed design can significantly increase device density on SiN chips and decreases the footprint of MZI by a factor of a least 50. This could help reduce the cost of low-loss high-performance photonic integrated circuits not only in telecommunications but also in other areas such as biosensors since SiN waveguides have a wide transparency window.

CHAPTER VI

REFLECTORS

Reflectors are necessary building blocks in integrated photonics that serve the same purpose as mirrors in free-space optics. A few examples of applications of reflectors can be found in filters, attenuators, and external cavity lasers (Karagodsky, Sedgwick, and Chang-Hasnain 2010). Sagnac loop mirrors (SLM) are one of the most basic types of reflectors. A SLM is simply a waveguide that loops back the traveling mode. In theory, a SLM can provide near a hundred percent reflection over a wide wavelength range, which is only limited by the dispersion and insertion losses of the couplers and the bending loss of the waveguide. However, for silicon nitride (SiN) waveguides (440 nm thick and 850 nm wide) cladded with silicon dioxide (SiO_2), where the minimum bending radiuses should be as high as 100 μm to minimize losses (Han, Ménard, and Shi 2020), SLMs can considerably increase the footprint of a device. Lowering the bending radius of SiN waveguides results in high losses and thus considerably lower the efficiency of the reflection.

6.1 Sagnac Loop Mirrors

A Sagnac loop mirror (SLM) is a basic reflector consisting of a looped waveguide connected to the output ports of a coupler. Figure 6-1 shows the schematic and the fabricated structure of this reflector where a bended waveguide of the radius of R is added to a MMI coupler. As the schematic shows, the loss in such a structure is mainly due to the coupler. However, the bending and propagation losses can be considerable

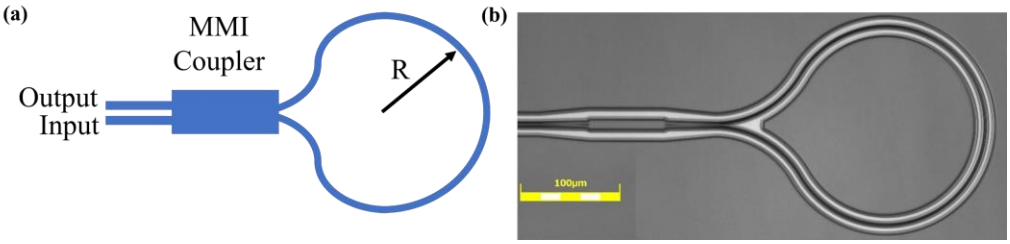


Figure 6-1: (a) Schematic and (b) image of fabricated structure of a Sagnac loop mirror using MMI couplers

depending on the radius of curvature. Nevertheless, there is a trade-off between the radius of curvature of the loop and the compactness of the reflector. Silicon waveguides can have a radius as low as 5 µm without any significant bending losses but SiN waveguides require much larger bending radius.

6.2 MZI Assisted Tunable Reflector

MZIs are among the most used devices in integrated photonics. The compact device demonstrated in chapter 5 can drastically reduce the total chip area required in complex or redundant systems such as wavelength division multiplexers (WDMs). We study the use of the proposed design in a tunable mirror, which can find applications in integrated lasers and transmitters (Tran *et al*. 2020; Lal *et al*. 2017), for instance. To implement this tunable mirror, a Sagnac loop mirror (SLM) is added at the output of MZI to reflect the optical energy towards the inputs of the coupler, as shown in Figure 6-2 (a). The fabricated structure based on this schematic is presented in Figure 6-3 (b). The radius of SLM should be at least 100 µm to insure low propagation losses and it is made of a waveguide that is 850 nm wide. For the MZI, we used the same configuration presented in Figure 5-1(a).

SLMs can provide high reflections if they have a radius of curvature large enough to minimize losses. Also, the propagation length inside the SLM is the same for both inputs since they propagate through the same length of waveguide but in opposite directions. Therefore, the addition of the SLM does not change the phase difference created by the MZI. The response of the mirror is completely controlled by the MZI since it sets the amount of light that is launched in the upper and lower branch of the SLM. For example, if the light at a given wavelength is split equally between the two ports of the SLM, then it will be transmitted to the output port. On the other hand, if all the light is incident at the upper branch of the SLM, then it will be reflected towards the input port.

Moreover, the light traversing this device goes through the MZI twice. Hence, to switch the state of the tunable mirror from a complete reflection of the light towards the input to total transmission toward the output requires only half of the phase change needed in a MZI switch (Soltani, Menard, and Kirk 2015). As demonstrated in the

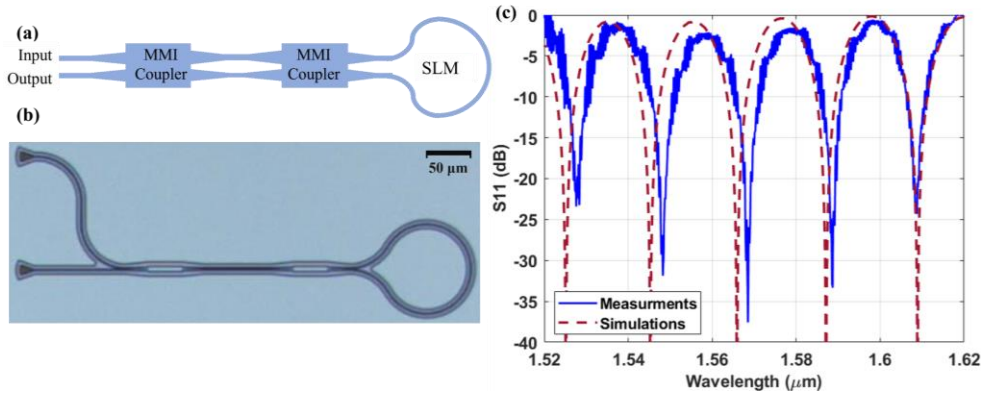


Figure 6-2: (a) Schematics of a tunable mirror using the proposed MZI and (b) image of the fabricated devices. (c) measurement results for the fabricated structure using a waveguide length of 150 µm. The dashed line shows the simulation results for the same structure.

previous section, this relative phase change can easily be generated by adding a local heater over the arms of the MZI. In addition, increasing the length of the MZI arms would further decrease the amount of power needed to switch state at the cost of reducing the FSR of the mirror.

The experimental results obtained from this structure are presented in Figure 6-2 (c) showing a FSR of 20.16 nm. Compared to a standalone MZI, the mirror has a FSR close to half that of the MZI (42.83 nm) because the MZI is in a folded configuration. Since the mode must travel twice through each MMI coupler, the insertion loss has increased to 2.56 dB. Simulation results are also plotted in this figure (dashed line). They predict a slightly larger FSR (21.39 nm) than what is measured from the fabricated device. This difference can be explained by variations of the effective and group index caused by the fabrication process.

6.3 Integrated Surface Grating Reflector

An alternative to SLMs that is often used in integrated circuits is the grating reflector. In search of gratings with wider bands and better reflections, many researchers focused on designs that use partially etched waveguides (Sharma *et al*. 2021) or novel optimization algorithms for obtaining improved structures (Yu, Cui, and Sun 2017). Although theoretically the results are fascinating, the proposed structures are either sensitive to fabrication errors or have complicated fabrication processes, and thus, have low fabrication yield. Fully etched circular gratings, such as those proposed in (Gao et al. 2016) and (Wang et al. 2017) on silicon-on-insulator (SOI) platforms, have less complicated design and fabrication processes. Based on these structures, we designed and investigated a circular Bragg grating reflector for SiN platforms that can be a

suitable substitution to Sagnac loop mirrors as a compact and wideband reflector. This is significant since we demonstrate that despite the lower index contrast on SiN platforms than on SOI, it is possible to create efficient and compact reflectors that are remarkably smaller than SLMs.

Figure 6-3 (a) shows a schematic of the circular grating where a waveguide of width w is connected to a taper right before the grating which has a period of Λ and a corrugation width of w_s. The reflection mechanism of this grating structure is based on the Fresnel diffraction principle. The width of the waveguide is small in comparison to the tapered area and thus the propagation of the light mode from the waveguide on the tapered area is the same as passing through a small aperture. The angle θ directly affects the diffracted light and the efficiency of the reflected power. Here we designed a reflector for the optical L band.

The inset of Figure 6-3 (a) illustrates the dimensions of the proposed grating reflector in comparison to a SLM designed for SiN for the same band and with a

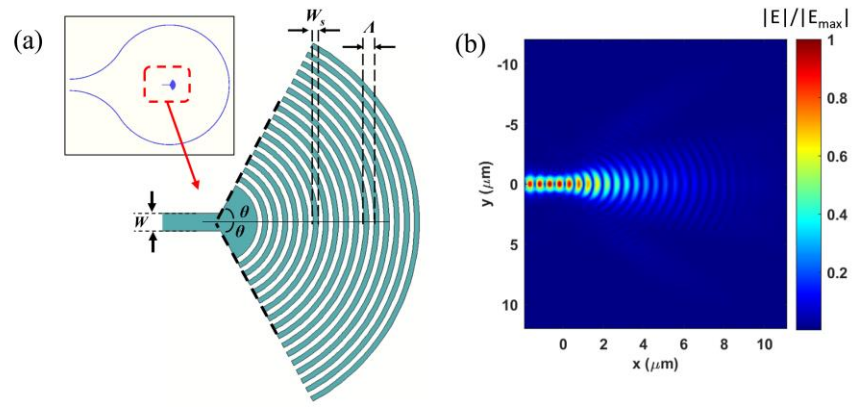


Figure 6-3: (a) Schematic of the grating reflector and its relative size in comparison to a SLM with a bending radius of 100 μm, (b) Normalized electric field distribution at a wavelength of 1595 nm

bending radius of 100 μm. The waveguide considered for the SLM, is 440 nm thick and 850 nm wide. Figure 6-3 (b) shows the distribution of the magnitude of the electric field inside the SiN waveguide, taper, and the grating at a wavelength of 1.59 μm for the TE polarization. This plot clearly illustrates the circular wavefront of the diffracted optical waves and the resulting focal point. The reflected light couples to and propagates back in the SiN waveguide.

The electric field presented in Figure 6-3 (b) is obtained with the 3D finite-difference time-domain (FDTD) simulation algorithm with Lumerical FDTD software. A full 3D simulation provides more accurate results for grating structures since it considers scattering losses from the corrugations in the vertical direction.

In general, the period of the corrugation can be calculated as $\Lambda = \lambda/(2n_{eff})$. An acceptable estimation for n_{eff} in this equation is (Magnusson and Shin 2003):

$$n_{eff} = n_c \cdot d + n_o(1 - d) \tag{6-1}$$

where d is the duty cycle of the gratings ($d = w_s/\Lambda$), n_c is the refractive index of the cladding oxide and n_o is the refractive index of the waveguide cores.

The reflector was designed to be implemented with a 450 nm thick silicon-rich SiN layer with a refractive index of 2.03 at a wavelength of 1.59 μm and cladded with 3.4 μm of thermal SiO_2 on the bottom and with the same thickness of plasma-enhanced chemical vapor deposited SiO_2 on the top. Owing to the lower index contrast of SiN-based optical stacks, the proposed grating period pitch Λ of 475 nm can be resolved by UV stepper lithography with sub-resolution assisted features (SRAF) (Cauchon *et al*. 2021) or electron beam lithography. Through numerical simulations, the optimum reflection was obtained with a duty cycle of 0.5. The width of the waveguide, *w*, is 850

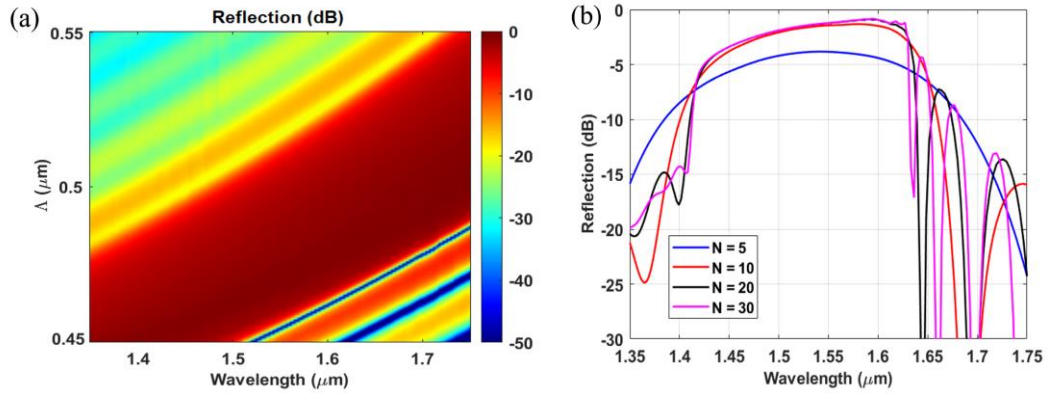


Figure 6-4: (a) Reflection spectrum of the grating for different periods as the function of wavelength, and (b) reflection for a period of 475 nm and different numbers of corrugations.

nm, which guarantees a single mode waveguide in both the C and L optical bands. Other important parameters controlling the reflections are the taper angle θ, the length of the tapered area, and the number of the corrugations, which all are optimized for the maximum reflection in the optical L band using simulation sweeps. The optimum values are 61.50^{o} for θ, 1.8 μm for the taper length, and 16 for the number of corrugations.

Figure 6-4 presents the impact of the period width Λ, and the number of corrugation N, on the reflection spectrum. Beyond a certain value for N, only the shape of the spectrum is changed as it gets sharper on the edge of the refection band without increasing the reflected power. By using the optimized values mentioned above, the minimum insertion loss is 0.82 dB and occurs at a wavelength of 1.59 μm. The 1- and 3-dB bandwidths are 130 and 195 nm, respectively. The bandwidth of the reflector is limited by the dispersive nature of this structure and scattering losses. These numbers were obtained with a structure that is at least 60 times smaller than a SLM and thus, has a significantly reduced footprint. The reflector fits in an area that is 25 by 20 μm.

In this chapter, we demonstrated how the proposed compact MZI in chapter 5 can be used as a tunable integrated mirror by adding a Sagnac loop. Similarly to the compact MZI, by using a heater above the MZI arms, the response of the mirror can be controlled and tuned. The experimental measurements are in agreement with simulations results. We also presented a Bragg circular grating reflector that could be a compact replacement for Sagnac loops mirrors in SiN platforms. The reflector was optimized using FDTD simulations to maximize reflection for TE mode in the L band. Although this structure presents a considerably lower footprint, the trade-off between the insertion loss and size should be carefully considered when using them as a substitution to SLMs.

CHAPTER VII

DESIGN, FABRICATION, AND MEASUREMENT OF FILTER STRUCTURES

A basic Fabry-Perot cavity consists of two mirrors separated by a distance *L*. The light that enters the cavity will go through multiple reflections between the mirrors and the interference between these reflections build up the cavity modes. We propose three different designs for Fabry-Perot based comb filters. The first two designs both use MZI assisted SLMs and the main difference between them is the structure of the MZI. The last design also uses integrated Bragg grating reflectors and the required phase difference is maintained by an extra length of waveguide added to one of the reflectors. In the first two designs, the FSR of the filter is dictated by the SLM waveguide length which is restricted by the bending losses of the loops, while the last design does not face such restriction. The three designs will be introduced and investigated after a brief review of the theory of these types of filters.

7.1 Analytical Study of the Filters

To build up the reflectors of the FP cavity, one can use MZI assisted SLMs, where a MZI is added before each reflector to create the phase difference required to control their reflectivity. This reflector has been introduced and analyzed in chapter 6. The FSR of the filter is determined by the overall length of the cavity, which corresponds to the propagation distance of the waves through the MZI arms, SLMs, and the waveguide connecting the two reflectors. To obtain large FSRs, the overall length of

the cavity should be short. Although SLMs are considered robust and wavelength independent structures, their large bending radius limits this design.

An alternative reflecting mechanism is to use broadband Bragg reflectors to replace the MZI assisted SLMs. Thus, instead of an MZI followed by a SLM, two Bragg grating reflectors are connected to the outputs of the MMI coupler. However, to obtain the required phase difference between the reflected waves, the two reflectors are connected to the MMIs with different waveguide lengths.

7.1.1 Reflection Mechanisms

We studied two different reflection mechanisms for the mirrors of the Fabry-Perot cavity which are the MZI assisted SLM, and the wide bands Bragg gratings. Figure 7-1 studies the first mirror response as a stand-alone structure shown in Figure 7-1 (a). To implement the structure in Lumerical INTERCONNECT, an ideal MZI block is used with a length difference of 0.3 μm. The plot presented in Figure 7-1(b) is controlled by the length difference of the MZI. Thus, the mirror response is independent of the radius

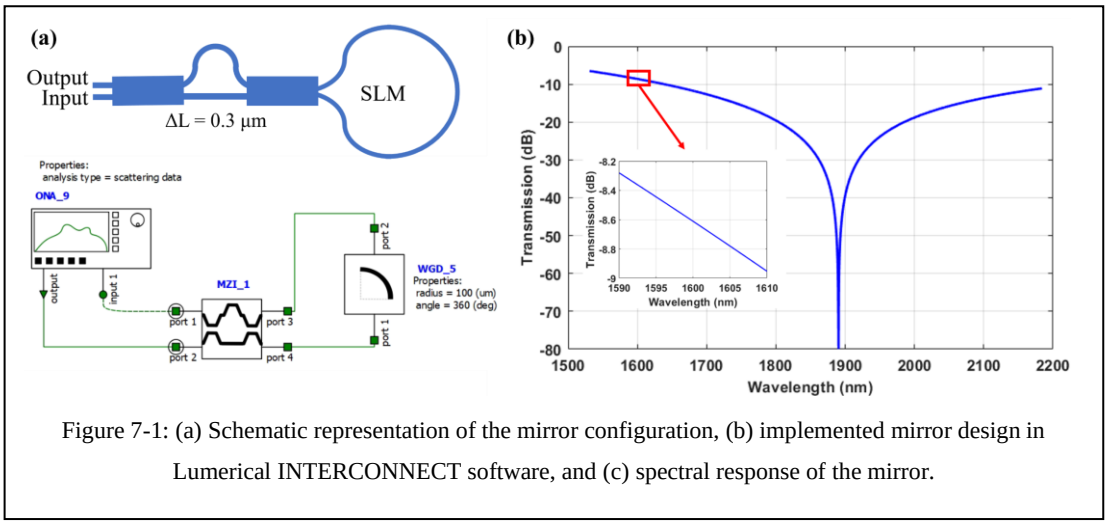


Figure 7-1: (a) Schematic representation of the mirror configuration, (b) implemented mirror design in Lumerical INTERCONNECT software, and (c) spectral response of the mirror.

of curvature of the Sagnac loop. The general filter response on the other hand is highly dependent on the length of the Sagnac loop since it corresponds to the cavity length.

The second configuration of the mirrors is presented as a stand-alone structure in Figure 7-2 (a). The reflection is achieved using Bragg reflectors after a 3-dB MMI coupler. The extra waveguide length added in front of one of the gratings in the mirrors (L_φ) controls the reflectivity of the mirror by adjusting the phase of the signals interfering at the output of the MMI.

To study the mirror response, the configuration is shown in Figure 7-2 (b) is implemented in Lumerical INTERCONNECT with ideal reflecting components instead of Bragg reflectors. The result is presented in Figure 7-2 (c). The inset in this figure shows the response at the L band. As mentioned above, the waveguide length L_φ introduces a phase difference between the reflected signals of the two gratings and results in an interferometric response, which sets the reflectivity of the mirror.

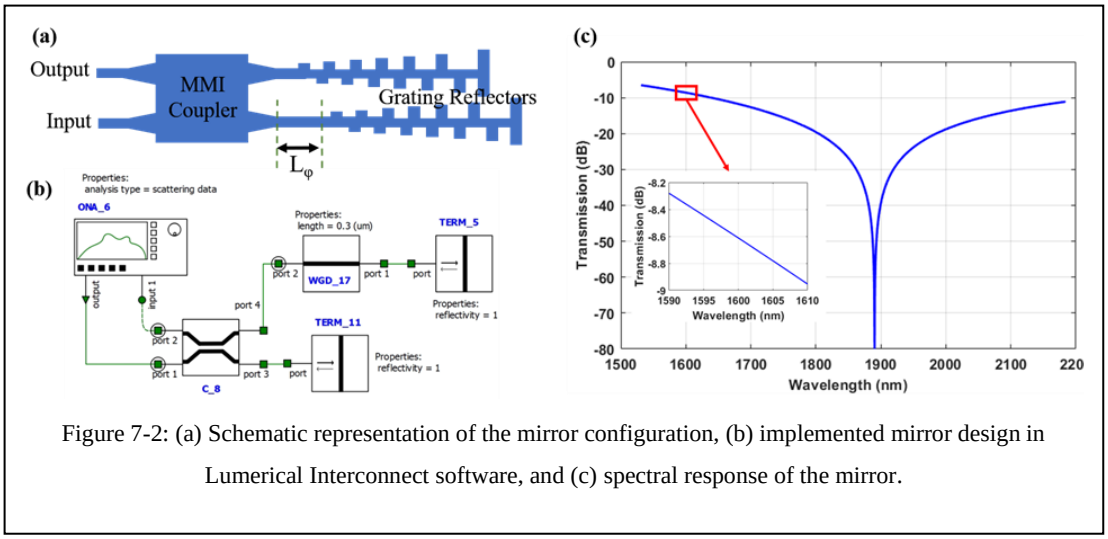


Figure 7-2: (a) Schematic representation of the mirror configuration, (b) implemented mirror design in Lumerical Interconnect software, and (c) spectral response of the mirror.

7.1.2 General Transfer Functions

One can calculate the transfer function of this device using the transfer matrix method (Saleh and Teich 2007). Figure 7-3 (a) shows one of the reflector arms of the filter with the propagation path of the light and the transfer matrix of each component. As shown here, a part of the light entering the reflector arm will be reflected as r_r while the rest is transmitted as t_r. Considering l_1, l_2, and l_3 as the lengths of different waveguides, the transmission factor of these waveguides can be described as:

$$a_m = exp\left(-\alpha l_m - i2\pi \frac{n_g l_m}{\lambda}\right) \qquad m = 1,2,3,4 \tag{7-1}$$

Where n_g is the group index and α is the loss factor. In the transfer matrix of the MMI, the factors κ and t are the splitting ratios of the couplers and are related to each other by $\kappa^2 + t^2 = 1$. The overall transfer matrix of a reflector arm is obtained by multiplying the matrices shown in Figure 7-3 which results in the following reflection and transmission factors

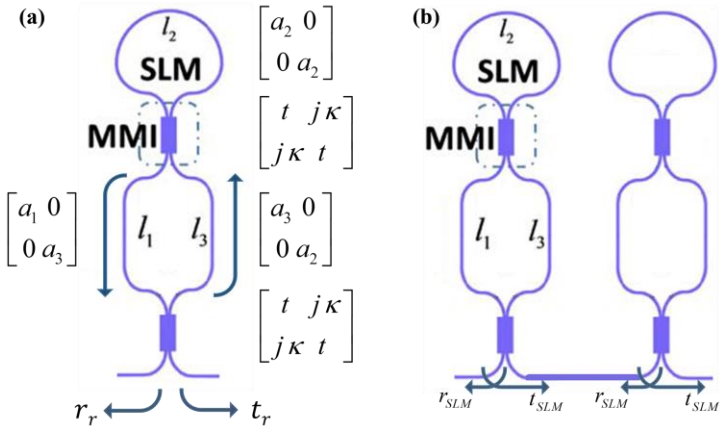


Figure 7-3: a) Schematic of the one reflector of the comb filter with the transfer matrix of each component and b) schematic of the complete filter design

$$r_r = j2a_2(a_1 + a_3)(a_1\kappa t^3 + a_3 t\kappa^3) \quad (7\text{-}2)$$

$$t_r = a_1 a_2 a_3(\kappa^4 + t^4) - 2a_2(a_1^2 + a_1 a_3 + a_3^2)t^2\kappa^2 \quad (7\text{-}3)$$

As shown in Figure 7-3 the total transmission function of the filter is the result of infinite series of reflection and transmission between the two reflector arms. Considering a_4 as the transmission factor of the waveguide between the two arms (cavity waveguide), the total transmission of the filter is:

$$T_{FP} = t_r a_4 t_r + t_r a_4 r_r a_4 r_r t_r + \ldots \quad (7\text{-}4)$$

This equation can be considered as a geometric series with the coefficient of $a_4 t_r^2$ and common ratio of $a_4^2 r_r^2$ and can be written as

$$T_{FP} = \frac{a_4 t_r^2}{1 - a_4^2 r_r^2} \quad (7\text{-}5)$$

Using this analytic equation one can estimate the output spectrum and FSR of the filter for the specific waveguide length of the MZI or the radius of curvature for the Sagnac loops.

For an ideal 3-dB MMI coupler, the splitting ratio between the two outputs is equal and therefore $\kappa = t = 0.707$. For a SiN waveguide that has a width of 850 nm and thickness of 450 nm, the measured group index is 2.1325. To have a comb filter, it is required to maintain a very small length difference (Δl) between the two arms of the MZI. Figure 7-4 (a) shows the calculated output transmission of the filter with a 1 μm length difference between the two arms of the MZI. Although the extinction ratio of the spectrum is mainly controlled by the splitting ratio of the couplers, it can be shown that even a 5% deviation from the 0.5 splitting ratio will only reduce the extinction ratio by 0.56 dB. Therefore, the designed and fabricated MMIs which were studied in chapter 5 are suitable for this filter structure.

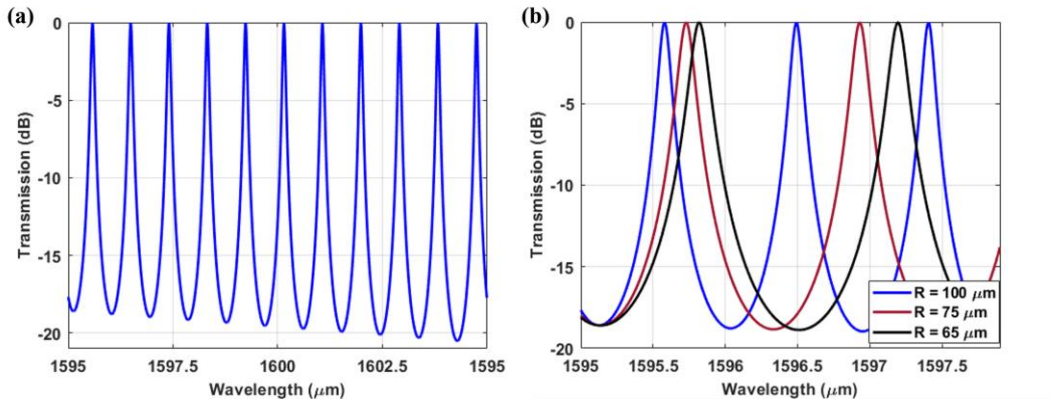


Figure 7-4: a) Calculated output spectrum of a filter with a Δl of 1 μm and radius of curvature of 75 μm for the SLMs and b) effect of the radius of curvature on the FSR and quality factor of the filter response.

The effect of the radius of the SLMs is studied in Figure 7-4 (b) where the output spectrum is plotted for three different radiuses. Decreasing the radius of the curvature will decrease the overall traveling length of the waves through the cavity and therefore, results in a larger FSR. This also affects the quality factor of the response as the resonances are sharper for smaller radiuses.

In the case that MZI assisted SLMs are replaced by Bragg gratings, the general mechanism of the filter cavity does not change. To use the transfer matrix method, one needs to consider the schematics in Figure 7-5. The mirror now consists of two Bragg grating reflectors connected to a MMI coupler with different waveguide lengths. A_m shows the reflectivity of the gratings and thus the light reflected toward the MMI coupler can be obtained from the following equations:

$$\begin{cases} A_2^- = A_m e^{-j2\beta l_\varphi} \\ A_3^- = A_m \end{cases} \tag{7-6}$$

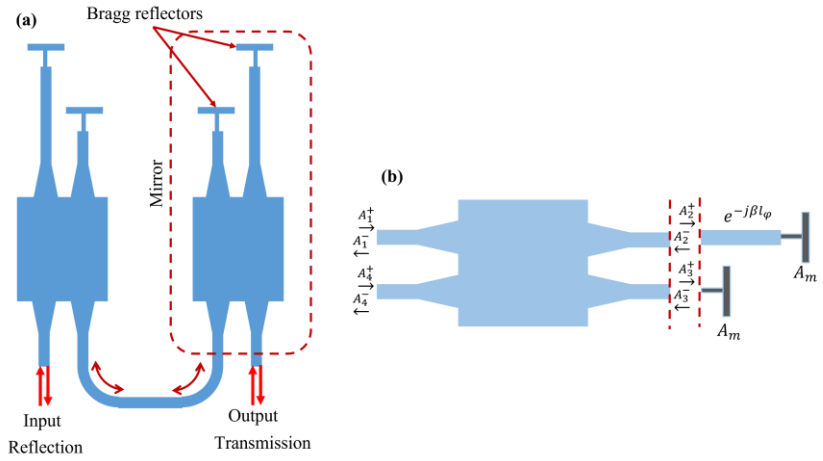


Figure 7-5: a) Schematic of the filter using Bragg gratings as reflecting mechanism of the cavity b) the inputs and outputs functions at ports of the MMI coupler and mirrors

where, as before, β is the propagation constant and the l_φ indicates the length difference between the two grating reflectors. Using the same coupler function as shown in Figure 7-3:

$$\begin{cases} A_2^+ = tA_1^+ + j\kappa A_4^+ \\ A_3^+ = j\kappa A_1^+ + tA_4^+ \end{cases} \tag{7-7}$$

$$\begin{cases} A_1^- = tA_2^- + j\kappa A_3^- \\ A_4^- = j\kappa A_2^- + tA_3^- \end{cases} \tag{7-8}$$

Therefore, the reflection (A_1^-) and transmission (A_4^-) of the mirror is obtained as

$$\begin{cases} A_1^- = \left(t^2 e^{-j2\beta l_\varphi} - \kappa^2\right) A_m A_1^+ + j\kappa t\left(e^{-j2\beta l_\varphi} - 1\right) A_4^+ \\ A_4^- = \left(t^2 e^{-j2\beta l_\varphi} - \kappa^2\right) A_m A_1^+ + j\kappa t A_m (e^{-j2\beta l_\varphi} - 1) A_4^+ \end{cases} \tag{7-9}$$

Considering these equations, the reflection and transmission factor of each mirror are:

$$r_r = \left(t^2 e^{-j2\beta l_\varphi} - \kappa^2\right) A_m \tag{7-10}$$

$$t_r = \left(t^2 e^{-j2\beta l_\varphi} - \kappa^2\right) A_m \tag{7-11}$$

Again, the total transfer function of the filter is built by geometrical series resulting in the following function:

$$T_{FP} = \frac{t_r e^{-j\beta l_{wg}}}{1 - r_r^2 e^{-2j\beta l_{wg}}} \tag{7-12}$$

Where l_{wg} is the length of the waveguide between the two mirrors which is also the length of the cavity. Using this analytic equation one can estimate the output spectrum and FSR of the filter for the specific waveguide lengths. Figure 7-6 shows the calculated response of the filter when the length of the waveguide cavity is 175 μm. The calculations show that as expected by increasing the cavity length smaller FSR and sharper resonances are achieved. In these calculations, the length difference of the waveguides connecting the Bragg gratings to the MMI couplers is considered as 0.3 μm.

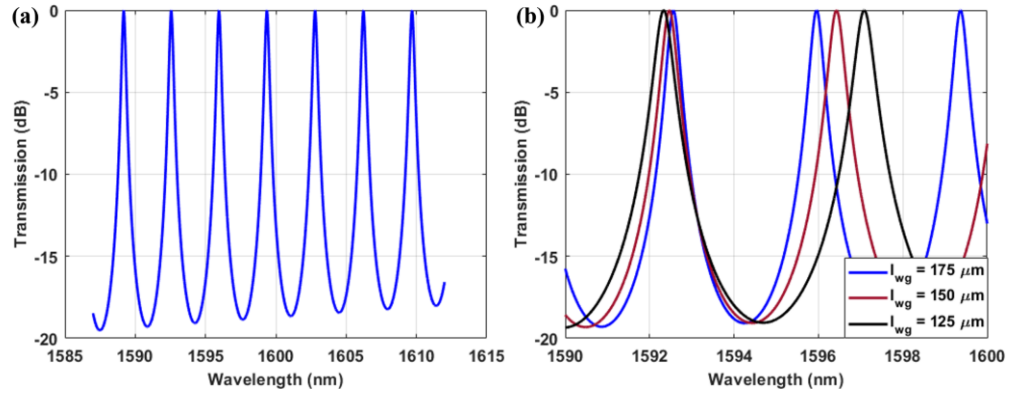


Figure 7-6: a) Output spectrum of a filter with l_φ = 0.3 μm l_{wg} of 175 μm and b) effect of length of the cavity on the FSR and quality factor of the filter response.

7.2 Bended Mach-Zehnder Assisted Sagnac Loop Mirror Filter

In this section, we demonstrate an easily fabricable SiN based comb filter with a small footprint. In this design, both the central wavelength and bandwidth can be tuned.

Figure 7-7 (a) shows a schematic of this resonator. Each SLMs is fed through a Mach-Zehnder interferometer (MZI) with multimode interferometer (MMI) couplers. The input signal first enters a 3-dB MMI coupler and is split into two similar beams passing through the MZI arms which have a slight difference in length. As a result of this length difference, the two beams are no longer in phase. These beams are then fed into the SLM by another 3-dB MMI coupler and then guided back through the MMI coupler, the MZI, and the input MMI coupler again. The output from this process passes along a waveguide and then enters the second loop experiencing the same procedure. This goes on multiple times (reflections and transmission from each loop) and results in a comb line output.

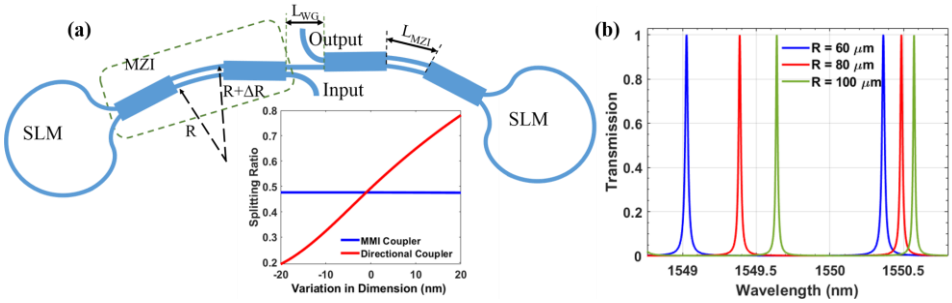


Figure 7-7: a) Schematic representation of the filter composed of two Sagnac loops and two MMI coupler assisted MZI. The inset shows a comparison of the splitting ratio of a MMI and a directional coupler as their dimension changes due to fabrication variations. b) Transmission spectrum of the filter for three different radiuses of the Sagnac loops showing how the length of the resonator affects the free spectral range and bandwidth.

The waveguides in this design are made of SiN and are 435 nm in width and 435 nm in height. To provide a symmetric propagation medium, the top and bottom cladding are both SiO_2. Using MMI couplers instead of conventional directional couplers (DC) lowers the sensitivity to fabrication errors, hence increasing yield. The inset of Figure 7-7 (a) demonstrates this by showing the variation in the splitting ratio of a SiN MMI and a DC of the same length when their dimensions change by ±20 nm. The MMI output hardly changes in this range whereas the output of the directional coupler changes severely.

Figure 7-7 (b) shows the transmission spectrum for three different radiuses of the SLMs. A quality factor of 7.75×10^4 is obtained for these designs when the propagation loss is 1.0 dB/cm. The length of the arms of the MZI was fixed at L_{MZI} = 80 μm with a small length difference of ΔL = 0.22 μm. This small length difference enables sharp resonances with a large extinction ratio. As ΔL increases, the bandwidth increases which results in a reduction of the extinction ratio. The length of the waveguide between the two loops, indicated by L_{WG} in Figure 7-7 (a), is 20 μm. As expected, reducing the lengths of the different sections in this resonator results in a wider free spectral range. However, to avoid bending losses, large radiuses are preferable for the SLMs. For the structure presented in Figure 7-7, the FSR is 1 nm when the bending radius is 80 μm.

To achieve a small length difference between the arms of the MZIs while preserving a bending radius of at least 80 μm, the arms are designed in such a way as to form two curves with a slightly different radius but with the same center of curvature. The architecture shown in Figure 7-7 (a) helps minimize propagation losses and the footprint of the filter. Considering a taper length of 30 μm at the inputs and outputs of

the MMI couplers to ensure efficient coupling, and a radius of 100 μm for the SLM to minimize bending losses, the overall dimension of the filter would not exceed 1300 μm by 245 μm, which corresponds to a footprint of 0.319 mm^2.

Although thermo-optical effects in silicon nitride are smaller than in silicon [4], they can still be used to tune the wavelength response of the device by adding a micro-heater on one arm of each MZIs. As shown in Figure 7-8 (a), the central wavelength of the transmission spectrum can be adjusted by manipulating the temperature on both MZIs. Figure 7-8 (b) shows how the wavelength response varies when the temperature changes in only one of the two MZIs. It results in a small shift in wavelength and an increase in the bandwidth at the cost of a reduction in transmission.

In this proposed design, the two Signac loop mirrors each fed through an unbalanced Mach-Zehnder interferometer create a resonant cavity. Using a large bending radius for the loop mirrors and a novel configuration for the interferometers

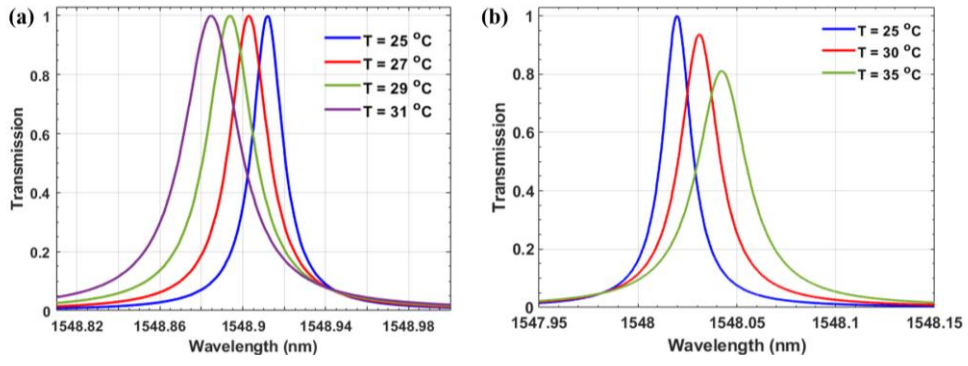


Figure 7-8: a) Transmission spectrum of the filter showing the thermo-optical effect when a) both MZIs and b) only one MZI arm are heated

that minimize the length of their arms and enables the implementation of a small length difference results in lower losses and footprint.

7.3 Compact Mach-Zehnder Assisted Sagnac Loop Mirror Filter

Although the Fabry - Perot filter discussed in the previous section is compact, the bended shape prevents efficient use of the floor space which results in the lower integration density of the devices. In this section, another realization of a compact comb filter is presented which is based on the MZI introduced in chapter 5. As fully explained before, the proposed MZI has a complete rectangle outline that makes it easy to effectively arrange the devices on the chip and increase the integration density.

The schematic of this filter which is shown in Figure 7-9 is similar to the previous design. When using SiN waveguides, the bending radius should be greater than 100 μm to ensure minimum losses. This restriction should be considered for all bended or curved waveguides in the device. This relatively high bending radius also affects the MZI design. Thus, in this design, we have used a MZI in which instead of two arms with different lengths, the arms had the same length but different widths (resulting in different effective refractive index).

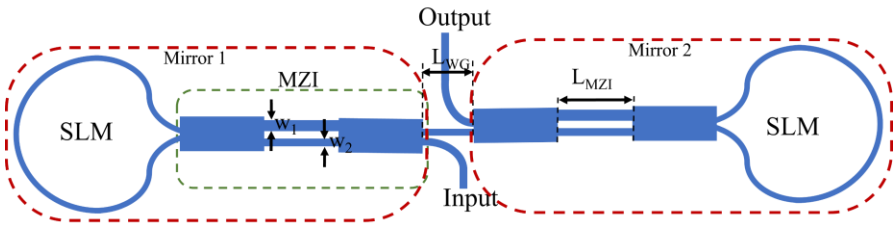


Figure 7-9: Schematic of the Fabry – Perot Filter with MZI assisted Sagnac loop mirror

To obtain the spectral response of the filter, the configuration shown in Figure 7-10 is implemented in Lumerical INTERCONNECT. For a more accurate simulation, the S-parameter model of the MMI coupler is exported from the Lumerical MODE software. Figure 7-11 (a) shows the response of the filter in which the SLMs have a radius of 100 μm, L_{wg} is 5 μm and the length of the MZI arms, L_{MZI} is 3 μm.

The spectrum shows 1.98 dB loss which is mainly due to the insertion losses of the MMI couplers. According to the fabrication results presented in previous chapters, the average insertion loss of each MMI is 0.42 dB, but the fact that optical wave will go through the mirrors multiple times explains higher total loss in the spectral response of the filter.

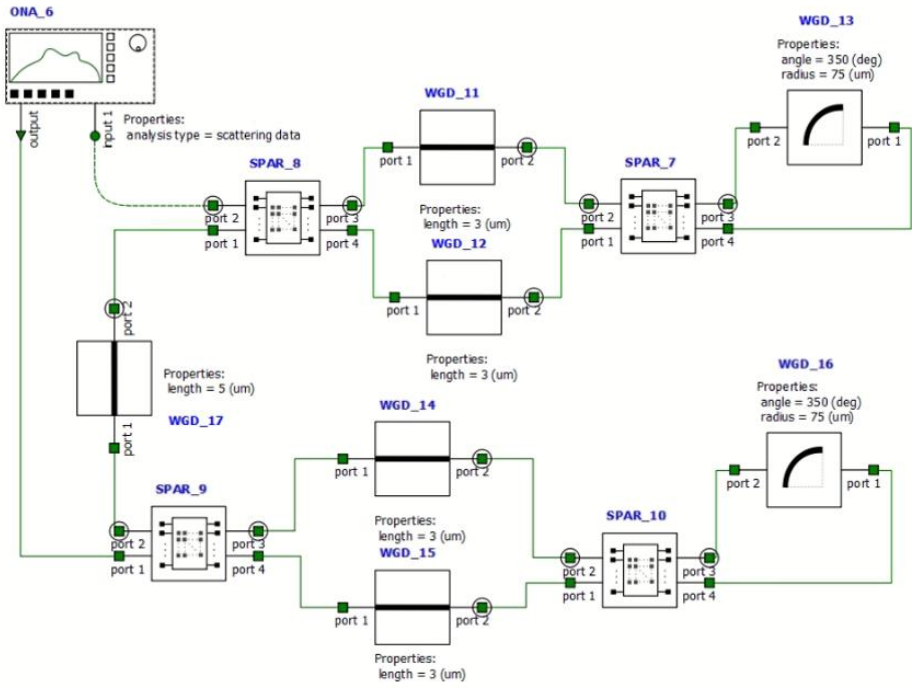


Figure 7-10: The filter model as implemented in Lumerical INTERCONNECT.

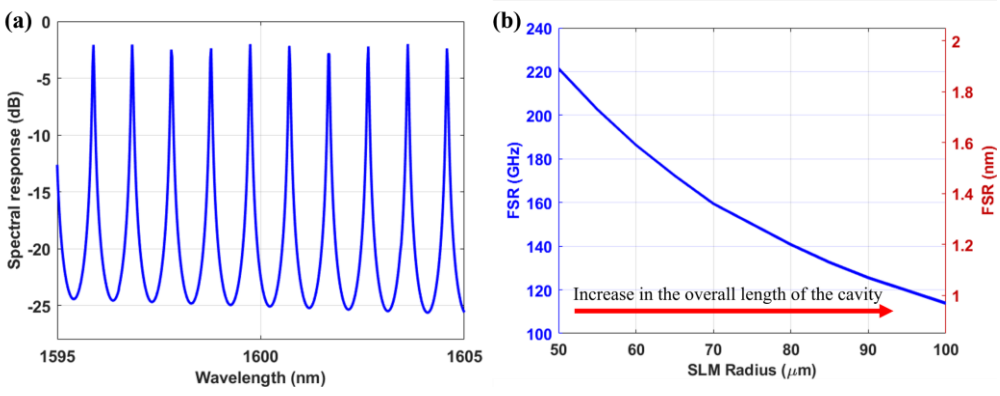


Figure 7-11: (a) Spectral response of the filter configuration with a SLM radius of 100 μm and (b) the FSR of the filter as a function of SLM radius.

The propagation loss of the optical mode in the waveguides is the second source of the losses in such filters. To analytically calculate the total insertion loss of a filter structure one can, modify equations (7-2) and (7-3) by adding the insertion loss of the couplers and choosing the right value for α in equation (7-1) to model the propagation losses. For the waveguide width of 850 nm, the reported average propagation loss from the measurements is 1.45 dB/cm in the L band center wavelength for the TE mode. The propagation loss is slightly lower for the TM mode (1.42 dB/cm) which suggests that a part of this loss is due to the sidewall roughness of the waveguides.

The last source of loss in this structure is the bending loss which is added to the system due to SLM. The averaged measured bending loss in the L band for each 90° bend with a radius of 100 μm is 0.027 dB in TE and 0.028 dB in TM mode. Lowering the bending radius is expected to increase this loss, but as Figure 7-11 (b) suggests and as explained before in this chapter, larger FSRs are only achieved by lowering the bending losses.

It has been shown by AEPONYX through measurements that by lowering the bending radius to 75 µm with a waveguide width of 850 nm, the average bending losses, do not show any significant difference than the curve of 100 µm in the L band. According to Figure 7-11 (b), this bending radius corresponds to a FSR of 153 GHz. The corresponding losses of smaller bending radiuses for a waveguide of 850 nm have not been measured.

Based on these measurements and simulations, we prepared three different filter designs for fabrication. The layouts are presented in Figure 7-12 (a). The main design had a SLM radius of 75 µm while the other two had SLM radiuses of 65 and 55 µm

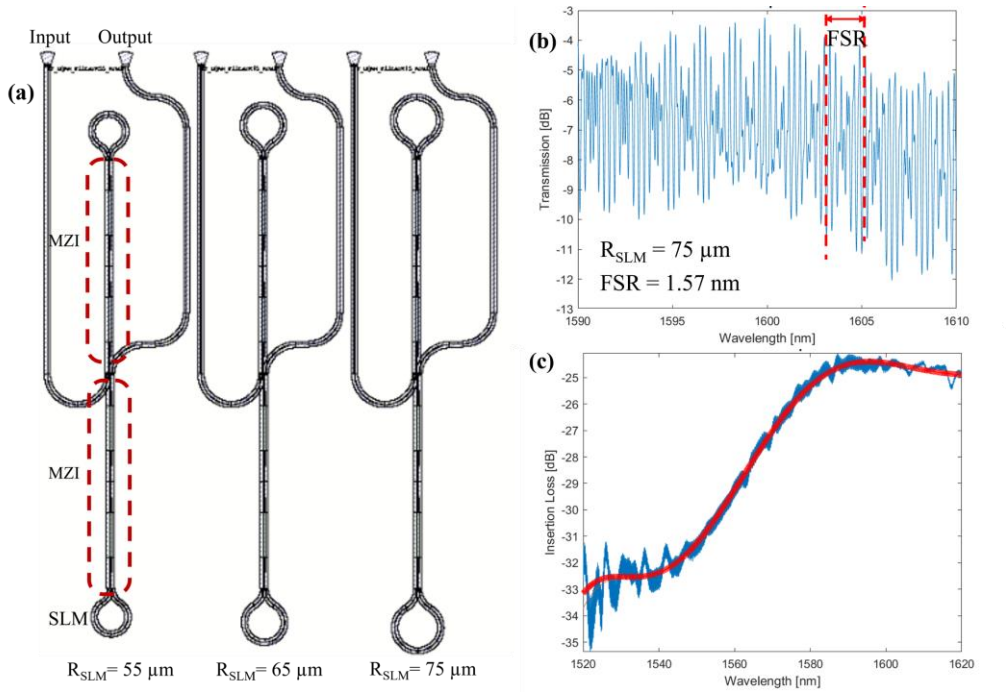


Figure 7-12: (a) Layout of the three designs with different SLM radiuses prepared for fabrication, (b) an example of the measure spectral response of the filter with a SLM radius of 75 µm at its output, and (c) response of the loopback reference structure in the same wavelength range.

and were expected to have a FSR of 153, 178, and 200 GHz, respectively. However, the measurement results of this fabrication were not as expected. Figure 7-12 (b) shows an example of the measured spectral response of the main design. As this figure shows, although the comb shape response is not obtained, there is a pattern of beating in the envelope of the response which is equal to the corresponding FSR of the filters. The same pattern is followed by the other two designs.

One possible explanation for these results can be the high loss of different components especially the MMI couplers. This filter is a Fabry-Perot cavity in which the mode travels through the structure multiple times and thus the loss of every component is amplified. The high noise of input grating couplers used in this fabrication made the situation worse. This can be observed from Figure 7-12 (c) which shows the response of the loopback structure.

To accurately study the reason for the difference between the results of simulations and measurements, another fabrication run is required in which the MMI couplers are added on the same die and close to the filter structure.

7.4 Bragg Grating Reflector Cavity Filter

The schematic presented in Figure 7-13 (a) is another implementation of a Fabry-Perot cavity in integrated optics. The set of the components enclosed with the red dashed lines acts as a mirror for the Fabry-Perot cavity, which was studied earlier in this chapter. The reflection is achieved using Bragg reflectors after a 3-dB MMI coupler. However, without the length difference of L_{φ} between the two reflectors, the destructive interference of the transmitted waves results in zero output from each mirror.

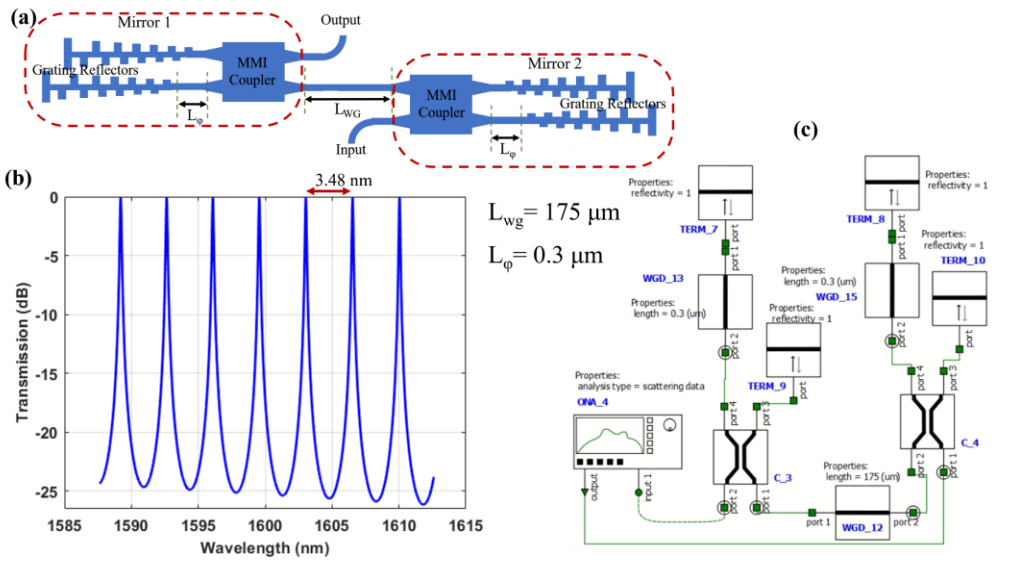


Figure 7-13: (a) Schematic of the second filter design using the compact MZI and grating reflectors. (b) Transmission spectrum of the filter, and (c) the filter model in Lumerical INTERCONNECT.

Note that the wavelengths that are not transmitted are reflected towards the input of the filter. Figure 7-13 (b) shows the spectral response of the filter representing a 400 GHz FSR. As Figure 7-13 (c) shows, at this stage, the results are obtained using ideal reflectors implemented in Lumerical Interconnect.

The proposed structure is tunable in wavelength using the thermo-optic effect if a metallic heater is placed above the waveguide, L_{wg}. In this case, 3 heaters can be used for thermal tuning the structure. As mentioned above, the one on L_{wg} will control the center wavelength whereas those on the mirrors will control the reflectivity and hence the shape of the response. If more stages are included, 2 heaters per stage should be added.

The FSR of the filter is determined by the length of the waveguide cavity. Since L_{wg} is very large compared to L_{φ}, the length of the cavity can be approximated by L_{wg}. Figure 7-14 shows the dependence of the FSR on the waveguide length. As this figure demonstrates, depending on the fabrication constraints, an FSR of more than 1 THz is achievable. Thus, this device is suitable to select one channel among up to 10 different ones.

In this chapter, after a brief study of two different reflecting mechanism, we reviewed our three proposed filter configurations, which are all based on interference in a Fabry - Perot cavity. In such filters, the FSR depends on the length of the cavity. Thus, in the first two configuration where a MZI assisted SLM is used as the reflecting mechanism, the FSR is limited. To solve this issue, we have proposed using a Bragg grating reflector that eliminated the need for a large bended waveguide, which drastically reduces the length of the cavity. With such structure we can easily reach FSRs as high as 1000GHz.

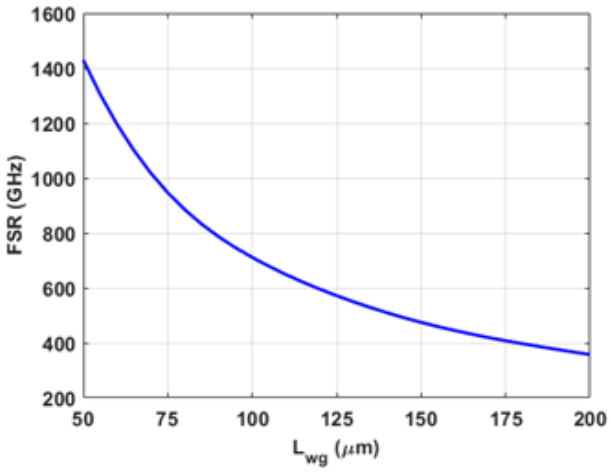


Figure 7-14: FSR of the filter as a function of the waveguide length.

CONCLUSION

In this thesis, three different filter configurations based on Fabry-Perot cavities in SiN platforms have been proposed. Before introducing the filter structures, we have designed, studied, and analyzed the different components used in their configurations. These components are MMI couplers, MZIs, and integrated reflectors. Because of their importance in the filter functionality, we have dedicated a chapter to each of these components.

In our designs, the MMI couplers replaced the commonly used directional couplers since they provide a better splitting ratio and fabrication tolerances. The 3 dB MMI couplers are designed and optimized to provide the lowest insertion loss and best coupling ratio. The simulation results are also confirmed by fabrications with the same lithography process which allow us to confirm the performance of the components prior to the final design. These results also confirms that as claimed the MMI couplers are not vulnerable to fabrication errors.

The 3 dB 2×2 MMI couplers that have been designed were then used to build a novel compact interferometer. In this interferometer, two arms have the same length but different width, which is in contrast to the conventional structure that has waveguide with the same width but different length. This means that the phase difference in the interference is obtained by the difference between the effective refractive indices instead than a length difference. This modification has reduced the required floor space of the MZI by a factor of at least 50. MZIs are essential components in many integrated optical devices and thus using the proposed design

instead of the conventional designs will result in an increase in the integration density which in turn reduces the size and costs of the systems on SiN platforms.

The proposed MZI is used in one of the three filter designs to reduce the size of the cavity and achieve the required FSR. The fabricated sample showed results that were in agreement with simulation and analytical calculations. We have also studied the feasibility of tuning the MZI response by using a local heater implemented on top of the MZI arms. It has been shown that by adding only one simple heater, the MZI spectrum can be tuned in order to compensate the fabrication errors. The analysis and study of this compact MZI is presented in an article submitted to the journal Optics Express after its concepts was introduced at the 2021 Frontiers in Optics and Laser Science conference (Armin, Nabki, and Ménard 2021a).

To built up the reflections in the cavities of filters on the SiN platform, we have investigated two different reflectors and one of them uses the proposed compact MZI and a Sagnac loop mirror. This MZI assisted SLM offers an FSR which is close to half that of a MZI with the same arm length and can be also tunable by using thermal effect induced by local heaters (This tunable reflector has been also included in our submission to the journal Optics Express). An alternative to this reflecting mechanism is integrated surface grating, which is also studied, analysed, and optimised for a wide band reflection spectrum. In comparison to the MZI assisted SLM grating reflectors are more compact and thus decrease the cavity length of the Fabry-Perot filter. Because of their smaller dimensions, the physical dimensions of the grating are more sensitive to fabrication errors and can result in a shift in spectrum. Nevertheless, the wide wavelength band of the reflector can compensate this effect. The new reflector design

and analysis has been presented at the 2021 IEEE Group Four Photonic Conference (Armin, Nabki, and Ménard 2021b).

After a careful study and the optimization of different filter components, we have reviewed the three proposed filter designs through analytical calculations as well as numerical simulations. In comparison to previous similar works (Zheng et al. 2017; Richard A. Soref, Leonardis, and Passaro 2018; Liu, Wu, and Dai 2019; Arianfard et al. 2021b; 2021a) the proposed designs are more compact, have lower propagation losses, are less sensitive to fabrication errors, and easy to tune.

The first two configurations use the MZI assisted SLMs, and the main difference is the MZI configurations. The first design uses a bended MZI in order to maintain the required small cavity length whereas the second one employs the proposed compact MZI. The new filter design proposed to use the bended MZI was presented at the 2020 OSA advanced photonics congress (Armin, Nabki, and Ménard 2020). Although both configurations succeed in reducing the cavity length and thus increasing the provided FSR, the total length is still limited by the bending radius of the SiN waveguides. The third proposed filter design is replacing the MZI assisted SLM with integrated gratings to overcome cavity length limitations. This substitution results in a filter configuration that can provide FSRs as high as 1000 GHz. The three different filter configurations presented here are compact structures with improved tolerance to fabrication errors and can provide high FSR. Thus, these designs successfully achieve objectives of this study as it was described in the first chapter.

This work proposed different solutions for some of the challenges of integrated comb filters in SiN platforms. Nevertheless, as expected, there are compromises. Using the bent MZI can reduce the cavity length but the bent shape of the filter results in less

efficient integration of the devices. Using the compact MZI solves this issue but results in a device which is more sensitive to fabrication tolerance since the performance of the MZI depends on the accuracy of the fabricated waveguides. The fabrication errors can be compensated by tuning the filter response using local heaters. However, the small length of the MZI arms in the filter structure create a new challenge for tuning which should also be addressed. The phase difference in the third filter configuration is obtained by the length of the waveguide which eliminates the need for an MZI. Yet it will be useful to study other grating configuration to overcome the fabrication challenge of the proposed designs.

REFERENCES

Abstreiter, Gerhard. 1992. “Engineering the Future of Electronics.” *Physics World* 5 (3): 36–40.

Alegria, Carlos A., Shalom S. Bergstein, Kimberly A. Dixson, and Christopher J. Hunt. 1999. “The WaveStar^{TM} BandWidth Manager: The Key Building Block in the next-Generation Transport Network.” *Bell Labs Technical Journal* 4 (1): 42–57.

Amatya, R., C. W. Holzwarth, M. A. Popović, F. Gan, H. I. Smith, F. Kärtner, and R. J. Ram. 2007. “Low Power Thermal Tuning of Second-Order Microring Resonators.” In , 5. Optical Society of America.

Arianfard, Hamed, Jiayang Wu, Saulius Juodkazis, and David J. Moss. 2021a. “Advanced Multi-Functional Integrated Photonic Filters Based on Coupled Sagnac Loop Reflectors.” *Journal of Lightwave Technology* 39 (5): 1400–1408.

———. 2021b. “High Performance Optical Filters Based on Advanced Coupled Sagnac Loop Waveguide Reflector Structures.” In *Silicon Photonics XVI*, 11691:17–24. SPIE.

Armin, Fahimeh, Frederic Nabki, and Michaël Ménard. 2020. “Fabrication Tolerant Compact Tunable Silicon Nitride Comb Filters.” In *OSA Advanced Photonics Congress (AP) 2020 (IPR, NP, NOMA, Networks, PVLED, PSC, SPPCom, SOF) (2020), Paper ITu4A.7*, ITu4A.7. Optical Society of America.

———. 2021a. “Compact Silicon Nitride Interferometer.” In *Frontiers in Optics + Laser Science 2021 (2021), Paper FTh6B.5*, FTh6B.5. Optica Publishing Group.

———. 2021b. “Compact Silicon Nitride Circular Grating Reflectors.” In *2021 IEEE 17th International Conference on Group IV Photonics (GFP)*, 1–2.

Baets, Roel, Roel Baets, Ananth Z. Subramanian, Ananth Z. Subramanian, Stéphane Clemmen, Stéphane Clemmen, Bart Kuyken, et al. 2016. “Silicon Photonics: Silicon Nitride versus Silicon-on-Insulator.” In *Optical Fiber Communication Conference (2016), Paper Th3J.1*, Th3J.1. Optical Society of America.

Blumenthal, Daniel J., René Heideman, Douwe Geuzebroek, Arne Leinse, and Chris Roeloffzen. 2018. “Silicon Nitride in Silicon Photonics.” *Proceedings of the IEEE* 106 (12): 2209–31.

Bryngdahl, Olof. 1973. “Image Formation Using Self-Imaging Techniques.” *JOSA* 63 (4): 416–19.

Cauchon, Jonathan, Jonathan St-Yves, Francois Menard, and Wei Shi. 2021. “Silicon Nitride Band Splitter Based on Multimode Bragg Gratings.” In *Optical Fiber Communication Conference (OFC) 2021 (2021), Paper F2B.5*. All virtual.

Cheben, Pavel, Robert Halir, Jens H. Schmid, Harry A. Atwater, and David R. Smith. 2018. “Subwavelength Integrated Photonics.” *Nature* 560 (7720): 565–72.

Chollet, Franck. 2016. “Devices Based on Co-Integrated MEMS Actuators and Optical Waveguide: A Review.” *Micromachines* 7 (2): 18.

Chrostowski, Lukas, and Michael Hochberg. 2015. *Silicon Photonics Design: From Devices to Systems*. Cambridge ; New York: Cambridge University Press.

“Cisco Annual Internet Report.” n.d. Cisco. https://www.cisco.com/c/en/us/solutions/executive-perspectives/annual-internet-report/index.html.

“Compound Annual Growth Rate.” 2021. In *Wikipedia*. https://en.wikipedia.org/wiki/Compound_annual_growth_rate.

Doylend, J. K., and A. P. Knights. 2012. “The Evolution of Silicon Photonics as an Enabling Technology for Optical Interconnection.” *Laser & Photonics Reviews* 6 (4): 504–25.

Du, Han, Fook Siong Chau, and Guangya Zhou. 2016. “Mechanically-Tunable Photonic Devices with On-Chip Integrated MEMS/NEMS Actuators.” *Micromachines* 7 (4): 69.

Essiambre, René Jean, and Robert W. Tkach. 2012. “Capacity Trends and Limits of Optical Communication Networks.” *Proceedings of the IEEE* 100 (5): 1035–55.

Gao, Shitao, Yang Wang, Ke Wang, and Efstratios Skafidas. 2016. “High Contrast Circular Grating Reflector on Silicon-on-Insulator Platform.” *Optics Letters* 41 (3): 520–23.

García-Meca, Carlos, Sergio Lechago, Antoine Brimont, Amadeu Griol, Sara Mas, Luis Sánchez, Laurent Bellieres, Nuria S Losilla, and Javier Martí. 2017. “On-Chip Wireless Silicon Photonics: From Reconfigurable Interconnects to Lab-on-Chip Devices.” *Light: Science & Applications* 6 (9): e17053–e17053.

Han, Qi, Michaël Ménard, and Wei Shi. 2020. “Superlattice Arrayed Waveguide Grating in Silicon Nitride.” *IEEE Photonics Technology Letters* 32 (22): 1411–14.

Jalali, Bahram, and Sasan Fathpour. 2006. “Silicon Photonics.” *Journal of Lightwave Technology* 24 (12): 4600–4615.

Karagodsky, Vadim, Forrest G. Sedgwick, and Connie J. Chang-Hasnain. 2010. “Theoretical Analysis of Subwavelength High Contrast Grating Reflectors.” *Optics Express* 18 (16): 16973–88.

Kish, Fred, Vikrant Lal, Peter Evans, Scott W. Corzine, Mehrdad Ziari, Tim Butrie, Mike Reffle, et al. 2018. “System-on-Chip Photonic Integrated Circuits.” *IEEE Journal of Selected Topics in Quantum Electronics* 24 (1): 1–20.

Komma, J., C. Schwarz, G. Hofmann, D. Heinert, and R. Nawrodt. 2012. “Thermo-Optic Coefficient of Silicon at 1550 Nm and Cryogenic Temperatures.” *Applied Physics Letters* 101 (4): 041905.

Kurdi, B. N., and D. G. Hall. 1988. “Optical Waveguides in Oxygen-Implanted Buried-Oxide Silicon-on-Insulator Structures.” *Optics Letters* 13 (2): 175–77.

Lal, Vikrant, Joseph Summers, Naksup Kim, Scott W. Corzine, Peter Evans, Matthias Lauermann, An Nguyen, et al. 2017. “Extended C-Band Tunable Multi-Channel InP-Based Coherent Transmitter PICs.” *Journal of Lightwave Technology* 35 (7): 1320–27.

Liu, Dajian, Hao Wu, and Daoxin Dai. 2019. “Silicon Multimode Waveguide Grating Filter at 2 Mm.” *Journal of Lightwave Technology* 37 (10): 2217–22.

Liu, Dajian, Hongnan Xu, Ying Tan, Yaocheng Shi, and Daoxin Dai. 2021. “Silicon Photonic Filters.” *Microwave and Optical Technology Letters* 63 (9): 2252–68.

Magnusson, R., and D. Shin. 2003. “Diffractive Optical Components.” In *Encyclopedia of Physical Science and Technology (Third Edition)*, edited by Robert A. Meyers, 421–40. New York: Academic Press.

Okamoto, Katsunari. 2005. *Fundamentals of Optical Waveguides*. 2nd edition. Amsterdam ; Boston: Academic Press.

Pruessner, Marcel W., Todd H. Stievater, Mike S. Ferraro, and William S. Rabinovich. 2007. “Thermo-Optic Tuning and Switching in SOI Waveguide Fabry-Perot Microcavities.” *Optics Express* 15 (12): 7557–63.

Rahim, Abdul, Eva Ryckeboer, Ananth Z. Subramanian, Stéphane Clemmen, Bart Kuyken, Ashim Dhakal, Ali Raza, et al. 2017. “Expanding the Silicon

Photonics Portfolio With Silicon Nitride Photonic Integrated Circuits." *Journal of Lightwave Technology* 35 (4): 639–49.

Reed, Graham T., and C. E. Jason Png. 2005. "Silicon Optical Modulators." *Materials Today* 8 (1): 40–50.

Ridder, R.M. de, K. Warhoff, A. Driessen, P.V. Lambeck, and H. Albers. 1998. "Silicon Oxynitride Planar Waveguiding Structures for Application in Optical Communication." *IEEE Journal of Selected Topics in Quantum Electronics* 4 (6): 930–37.

Saber, Md Ghulam, Nicolás Abadía, Yun Wang, and David V. Plant. 2018. "Fabry–Perot Resonators with Transverse Coupling on SOI Using Loop Mirrors." *Optics Communications* 415 (May): 121–26.

Saleh, Bahaa E. A., and Malvin Carl Teich. 2007. *Fundamentals of Photonics*. 2nd edition. Hoboken, N.J: Wiley-Interscience.

Schmidtchen, J., A. Splett, B. Schuppert, K. Petermann, and G. Burbach. 1991. "Low Loss Singlemode Optical Waveguides with Large Cross-Section in Silicon-on-Insulator." *Electronics Letters* 27 (16): 1486–88.

Sharma, Tarun, Vaibhav Rana, Jiaqi Wang, Zhenzhou Cheng, Yuzhi Chen, Youfu Geng, Xueming Hong, Xuejin Li, Kyoungsik Yu, and Brajesh Kumar Kaushik. 2021. "Design of Grating Based Narrow Band Reflector on SOI Waveguide." *Optik* 227 (February): 165995.

Shimogaki, Tetsuya, Hikaru Fukushima, Daisuke Inoue, Tadashi Ichikawa, and Tatsuya Yamashita. 2021. "Polarization Splitting Grating Coupler for Optical Antenna of LiDARs Consisting of Two Gratings Coupling Light in a Perfectly Vertical Direction." *IEEE Photonics Journal* 13 (3): 1–11.

Soldano, L.B., and E.C.M. Pennings. 1995. "Optical Multi-Mode Interference Devices Based on Self-Imaging: Principles and Applications." *Journal of Lightwave Technology* 13 (4): 615–27.

Soltani, Fatemeh, Michael Menard, and Andrew G. Kirk. 2015. "Optical Modulator with an Integrated Loop Mirror." In *2015 IEEE Optical Interconnects Conference (OI)*, 100–101.

Soref, R. A., and J. P. Lorenzo. 1985. "Single-Crystal Silicon: A New Material for 1.3 and 1.6 Mm Integrated-Optical Components." *Electronics Letters* 21 (21): 953–54.

Soref, R.A. 1993. “Silicon-Based Optoelectronics.” *Proceedings of the IEEE* 81 (12): 1687–1706.

Soref, Richard. 2006. “The Past, Present, and Future of Silicon Photonics.” *IEEE Journal of Selected Topics in Quantum Electronics* 12 (6): 1678–87.

Soref, Richard A., Francesco De Leonardis, and Vittorio M. N. Passaro. 2018. “Mach-Zehnder Crossbar Switching and Tunable Filtering Using N-Coupled Waveguide Bragg Resonators.” *Optics Express* 26 (12): 14959–71.

Suzuki, S., S. Sumida, Y. Inoue, M. Ishii, and Y. Ohmori. 1997. “Polarisation-Insensitive Arrayed-Waveguide Gratings Using Dopant-Rich Silica-Based Glass with Thermal Expansion Adjusted to Si Substrate.” *Electronics Letters* 33 (13): 1173.

Thomson, D. J., Y. Hu, G. T. Reed, and Jean-Marc Fedeli. 2010. “Low Loss MMI Couplers for High Performance MZI Modulators.” *IEEE Photonics Technology Letters* 22 (20): 1485–87.

Tran, Minh A., Duanni Huang, Joel Guo, Tin Komljenovic, Paul A. Morton, and John E. Bowers. 2020. “Ring-Resonator Based Widely-Tunable Narrow-Linewidth Si/InP Integrated Lasers.” *IEEE Journal of Selected Topics in Quantum Electronics* 26 (2): 1–14.

Trinh, P.D., S. Yegnanarayanan, F. Coppinger, and B. Jalali. 1997. “Silicon-on-Insulator (SOI) Phased-Array Wavelength Multi/Demultiplexer with Extremely Low-Polarization Sensitivity.” *IEEE Photonics Technology Letters* 9 (7): 940–42.

Wang, Yang, Shitao Gao, Ke Wang, Hongtao Li, and Efstratios Skafidas. 2017. “Ultra-Broadband, Compact, and High-Reflectivity Circular Bragg Grating Mirror Based on 220 Nm Silicon-on-Insulator Platform.” *Optics Express* 25 (6): 6653–63.

Winzer, Peter J., and David T. Neilson. 2017. “From Scaling Disparities to Integrated Parallelism: A Decathlon for a Decade.” *Journal of Lightwave Technology* 35 (5): 1099–1115.

Winzer, Peter J., David T. Neilson, and Andrew R. Chraplyvy. 2018. “Fiber-Optic Transmission and Networking: The Previous 20 and the next 20 Years [Invited].” *Optics Express* 26 (18): 24190–239.

Yu, Zejie, Haoran Cui, and Xiankai Sun. 2017. “Genetically Optimized On-Chip Wideband Ultracompact Reflectors and Fabry-Perot Cavities.” *Photonics Research* 5 (6): B15–19.

Zheng, Shuang, Nan Zhou, Yun Long, Zhengsen Ruan, Jing Du, Xiao Hu, Li Shen, Shuhui Li, and Jian Wang. 2017. “Compact Tunable Photonic Comb Filter on a Silicon Platform.” *Optics Letters* 42 (14): 2762–65.